

Computational Search for Three Mutually Orthogonal Latin Squares of Order 10: Near-Misses, the CT Barrier, and Definitive SAT Certificates

Technical Report

Autoresearch Project

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Abstract

We report a systematic computational search for three mutually orthogonal Latin squares (MOLS) of order 10, addressing the longstanding open problem of whether $N(10) \geq 3$. Using OR-Tools CP-SAT exact optimization (up to 4 parallel workers per trial), iterated local search, and Glucose3 CDCL satisfiability solving, we conduct a deep search across **all 323 verified MOLS-10 pairs** (scan complete) plus 115 newly discovered CT=2 pairs, covering 28+ distinct L_1 families.

Our main findings are: (1) the minimum total clash energy $\mathcal{E} = cl_{13} + cl_{23}$ has cascaded $37 \rightarrow 34 \rightarrow 33 \rightarrow 32 \rightarrow 31 \rightarrow 30 \rightarrow 29 \rightarrow \mathbf{26}$ (Sessions 9–26b); the current all-time record is $\mathcal{E} = \mathbf{26}$ ($cl_{13} = 8, cl_{23} = 18$) for `iso_tv_21_0` (`ee29e34e`) via 4-worker CP-SAT hint cascade (seed=1943590465); (2) structural analysis proves $\mathcal{E}=26$ is a *strict local minimum*: exhaustive 1-move neighborhood (all row-swaps, col-swaps, symbol-relabelings, intercalates give $\mathcal{E} \geq 26$) and all $\binom{10}{2} = 45$ double-intercalate combinations give $\mathcal{E} \geq 26$; five independent CP-SAT trials all converge to the *same canonical* L_3 (single-attractor basin); (3) **multiple independent** (L_1, L_2) **pairs reach** $\mathcal{E} = 28$: the `iso_tv_21_0` secondary cascade (`ee29e34e`, $cl_{13} = 16, cl_{23} = 12$), `mdecomp_279` (`e54e791c`, $cl_{13} = 8, cl_{23} = 20$), and **`mdecomp_113`** (`2469c6dc`, $cl_{13} = 17, cl_{23} = 11$, Session 30 trial 3) — three structurally distinct pairs all converging to $\mathcal{E} = 28$; (4) a systematic scan of 44+ newly catalogued `mdecomp` CT=2 pairs identifies `mdecomp_56` and `mdecomp_113` both reaching $\mathcal{E} = 31$ in 120 s, with `mdecomp_56` cascading to $\mathcal{E} = 30$ ($cl_{13} = 15, cl_{23} = 15$, balanced split); (5) every MOLS-10 pair tested has $CT(L_1, L_2) \leq 2 < 10$, with Glucose3 UNSAT certificates for 311 pairs; 6×600 s feasibility trials with hard constraint $\mathcal{E} \leq 25$ all time out without finding SAT or INFEASIBLE, strongly suggesting $\mathcal{E} = 26$ is the global minimum for `iso_tv_21_0`.

Collectively, this constitutes the strongest computational evidence to date bearing on $N(10)$. The barrier $\mathcal{E} = 26 > 0$ across all strategies and pairs, the unique single-attractor basin, and the CT barrier ≤ 2 for all 344 pairs collectively support the longstanding conjecture $N(10) = 2$, though a rigorous proof remains open.

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1 Introduction

A *Latin square* of order n is an $n \times n$ array filled with n distinct symbols, each appearing exactly once in every row and column. Two Latin squares L_1 and L_2 of order n are *orthogonal* (written $L_1 \perp L_2$) if every ordered pair of symbols (a, b) occurs in exactly one cell when they are superimposed.

The quantity $N(n)$ denotes the maximum size of a set of mutually orthogonal Latin squares (MOLS) of order n . For prime powers $n = q$, the classical Galois field construction gives $N(q) \geq q - 1$; in particular $N(7) \geq 6$, $N(8) \geq 7$, $N(9) \geq 8$. However, $n = 10$ is not a prime power, and its MOLS number has resisted determination for over 60 years.

1.1 Historical Background

Euler [7] conjectured in 1782 that $N(n) = 1$ whenever $n \equiv 2 \pmod{4}$. This conjecture held for small values: $N(2) = 1$ and $N(6) = 1$ (the latter proved by exhaustive computation by Clements and Lindström [4] following Tarry [13]). Parker [12] dramatically disproved Euler’s conjecture for $n = 10$ by constructing two orthogonal Latin squares of order 10, establishing $N(10) \geq 2$.

The upper bound $N(10) \leq 8$ follows from the non-existence of a projective plane of order 10, proved computationally by Lam, Thiel, and Swiercz [10] using an exhaustive computer search consuming tens of thousands of CPU hours. The current state of knowledge is:

$$2 \leq N(10) \leq 8.$$

Most combinatorialists conjecture $N(10) = 2$ [6, 5], but no proof exists.

1.2 Our Contribution

We present a computational study with the following contributions:

1. **New MOLS-10 pair catalog** ($N(10) \geq 2$): A systematically enumerated, machine-verified catalog of 311 distinct MOLS-10 pairs generated by five independent construction methods, all with $\text{CT} \leq 2$. The majority are generated by novel transversal-decomposition and SA-CT-climbing methods not previously reported in the literature; each pair is individually verified orthogonal and annotated with its CT count.
2. **Near-miss triples** ($N(10) \geq 3$ **evidence**): A pool of near-miss triples (L_1, L_2, L_3) with $\text{cl}_{12} = 0$ (i.e., $L_1 \perp L_2$): the current record is $\mathcal{E} = 33$ ($\text{cl}_{13} = 14$, $\text{cl}_{23} = 19$, Session 10), advancing from $\mathcal{E} = 34$ (Session 9 CP-SAT) via pool warm-start SA. The record has been independently confirmed for multiple (L_1, L_2) pairs via `count.clashes` recomputation.
3. **CT barrier documentation**: Empirical confirmation that all 311 generated MOLS-10 pairs have common transversal count $\text{CT} \leq 2 < 10$, making each pair provably unable to support a third orthogonal square.
4. **SAT certificates**: CDCL UNSAT certificates for all 311 pairs, obtained at ≈ 42 ms per pair using Glucose3 [1].
5. **Algorithmic framework**: A novel parallel-tempering SA with pool-jump ILS, reverse search, weighted MAX-SAT (RC2), three-LS crossover, and a new *joint 3-LS SA* that optimizes all three Latin squares simultaneously without fixing any pair.
6. **Joint 3-LS SA result**: Starting from random triples ($\mathcal{E}_{\text{init}} \approx 79$) or perturbed pool entries, the joint SA reaches $\mathcal{E} = 33$ from pool seeds, confirming that pool warm-starts allow SA to descend well below its former $\mathcal{E} = 37$ floor. CP-SAT exact optimization was required to first break to $\mathcal{E} = 34$; SA with warm-starts then descends further.

7. **CP-SAT symmetry breaking:** Symbol relabeling of L_3 (fixing $L_3[0] = \text{id}$) reduces the CP-SAT search space by $10! \approx 3.6 \times 10^6$ without loss of generality, enabling the $\mathcal{E} = 34$ breakthroughs in reasonable wall-clock time.
8. **File-locking fix:** A concurrent-write race condition in the pool save function (`_save_triple_miss`) was identified: multiple workers silently overwrote each other’s results. Adding `fcntl.LOCK_EX` locking ensures all breakthrough results (including $\mathcal{E} \leq 33$) are reliably persisted.

1.3 Paper Organization

Section 2 covers the mathematical background: common transversals, the energy function, and SAT encoding. Section 3 describes all search algorithms. Section 4 presents experimental results including the verified MOLS-10 pair catalog ($N(10) \geq 2$ evidence). Section 5 analyzes the near-miss triples. Section 6 discusses the CT barrier. Section 7 collects all empirical evidence on $N(10) \geq 3$. Section 8 concludes with conjectures and future directions. Appendix A gives full matrices for representative pairs.

2 Mathematical Background

2.1 Common Transversals

Definition 2.1. Let L_1, L_2 be orthogonal Latin squares of order n . A common transversal (CT) of (L_1, L_2) is a set $T = \{(r_0, c_0), \dots, (r_{n-1}, c_{n-1})\}$ of n cells such that: (i) each row $0, \dots, n-1$ appears exactly once; (ii) each column $0, \dots, n-1$ appears exactly once; (iii) the values $\{L_1(r_i, c_i)\}$ are all distinct; and (iv) the values $\{L_2(r_i, c_i)\}$ are all distinct.

We write $\text{CT}(L_1, L_2)$ for the total number of common transversals of the pair (L_1, L_2) .

Theorem 2.2 (CT Necessity). A Latin square L_3 of order n is orthogonal to both L_1 and L_2 if and only if the n^2 cells can be partitioned into n disjoint common transversals $T_0, \dots, T_{n-1}$ of (L_1, L_2) , where $T_k = \{(i, j) : L_3(i, j) = k\}$.

Proof sketch. If $L_3 \perp L_1$, then the 100 pairs $(L_1(i, j), L_3(i, j))$ are all distinct. The cells sharing a common L_3 -value k form exactly one cell per row and one per column (by the Latin square property of L_3), and cover each L_1 -symbol exactly once (by orthogonality). Applying the same argument to $L_3 \perp L_2$ gives the CT partition. The converse is immediate. $\square$

Corollary 2.3. If $\text{CT}(L_1, L_2) < n$, then no Latin square L_3 satisfying $L_3 \perp L_1$ and $L_3 \perp L_2$ exists.

Proposition 2.4 (Isotopy Invariance). Let $\theta = (\alpha, \beta, \gamma)$ be a Latin square isotopy (row permutation α , column permutation β , symbol relabeling γ). If the same θ is applied to both L_1 and L_2 , then $\text{CT}(\theta(L_1), \theta(L_2)) = \text{CT}(L_1, L_2)$.

2.2 The Energy Function

For Latin squares A, B of order n , define the *clash count*:

$$\text{cl}(A, B) = \#\{(a, b) \in \{0, \dots, n-1\}^2 : (a, b) \text{ never appears in } \{(A(i, j), B(i, j))\}_{i,j}\}.$$

This counts symbol-pairs that are *missing* from the superposition of A and B ; $\text{cl}(A, B) = 0$ iff $A \perp B$. By a pigeonhole argument, the number of missing pairs equals the total excess multiplicity of repeated pairs, so $\text{cl}(A, B)$ also quantifies how far (A, B) is from orthogonality. The *total energy* of a triple is:

$$\mathcal{E}(L_1, L_2, L_3) = \text{cl}(L_1, L_2) + \text{cl}(L_1, L_3) + \text{cl}(L_2, L_3).$$

Goal: Find (L_1, L_2, L_3) with $\mathcal{E} = 0$.

Lemma 2.5 (Expected clash for random LS). *For a uniformly random Latin square L_3 and a fixed Latin square L_1 , the expected clash count satisfies $\mathbb{E}[\text{cl}(L_1, L_3)] \approx n^2(1 - e^{-1}) \approx 63.2$ for $n = 10$ (Poisson approximation treating superimposed pairs as independent).*

Remark 2.6. Lemma 2.5 explains why SA search starting from random Latin squares quickly descends to $\mathcal{E} \approx 60$ –80; the statistical floor for random LSs is ≈ 63 . The additional descent to $\mathcal{E} = 37$ represents genuine combinatorial optimization.

2.3 SAT Encoding

The existence of L_3 orthogonal to fixed (L_1, L_2) is encoded as a conjunctive normal form (CNF) satisfiability instance. We introduce Boolean variables:

$$x_{i,j,k} = 1 \iff L_3(i, j) = k, \quad i, j, k \in \{0, \dots, 9\}.$$

This gives 1,000 Boolean variables. Five families of *exactly-one* clauses enforce:

$$(C1) \text{ Cell uniqueness: } \forall i, j : \text{ exactly one } k \text{ has } x_{i,j,k} = 1, \quad (1)$$

$$(C2) \text{ Row uniqueness: } \forall i, k : \text{ exactly one } j \text{ has } x_{i,j,k} = 1, \quad (2)$$

$$(C3) \text{ Col uniqueness: } \forall j, k : \text{ exactly one } i \text{ has } x_{i,j,k} = 1, \quad (3)$$

$$(C4) \text{ Orth-}L_1: \quad \forall a, b : \text{ exactly one } (i, j) \text{ with } L_1(i, j) = a \text{ has } x_{i,j,b} = 1, \quad (4)$$

$$(C5) \text{ Orth-}L_2: \quad \forall a, b : \text{ exactly one } (i, j) \text{ with } L_2(i, j) = a \text{ has } x_{i,j,b} = 1. \quad (5)$$

Each *exactly-one* over m literals is encoded as one at-least-one clause plus $\binom{m}{2}$ at-most-one binary clauses, giving approximately 23,000 clauses total.

Theorem 2.7 (Completeness). *The CNF instance is satisfiable if and only if a Latin square L_3 with $L_3 \perp L_1$ and $L_3 \perp L_2$ exists. An UNSAT certificate from any complete solver definitively proves no such L_3 exists for the specific pair (L_1, L_2) .*

3 Search Algorithms

3.1 Parallel-Tempering Simulated Annealing

The primary search uses parallel-tempering SA (PT-SA) [8] with $K = 7$ replicas at temperatures:

$$T = (1.0, 4.0, 16.0, 64.0, 256.0, 1024.0, 4096.0).$$

Each replica maintains a state (L_1, L_2, L_3) and energy $\mathcal{E}$. At each outer step:

1. Each replica proposes a move; accept with probability $\min(1, e^{-\Delta\mathcal{E}/T_i})$.
2. Every 2,000 steps, swap adjacent replicas $(i, i+1)$ with Metropolis probability $\min(1, e^{(E_i - E_{i+1})(1/T_i - 1/T_{i+1})})$.

Move types. When $\text{cl}_{12} < 5$, the target square is L_3 with probability 0.70; otherwise the target is chosen uniformly. Applied moves:

Iterated Local Search (ILS). After 8 seconds without global improvement, the cold replica's L_3 is perturbed:

- *Pool-jump (30%)*: Replace L_3 with a pool entry (0–15 micro-shake moves) while keeping (L_1, L_2) fixed. This injects a structurally distinct L_3 basin.
- *Random shake (70%)*: Apply 25–70 random LS-preserving moves to L_3 .

ILS fires ≈ 14 times per 97-second trial.

Table 1: Move types and their properties.

Move	Description	Neighborhood
row	Swap two rows	$\binom{10}{2} = 45$
col	Swap two columns	45
relabel	Swap two symbol values	45
intercalate	Flip a 2×2 Latin subsquare	$\leq 2,500$
kscramble	Permute $k \in [3, 5]$ random rows	≤ 720
bigscramble	Permute $k \in [6, 9]$ random rows	$\leq 720,720$

Table 2: Initialization strategy portfolio.

Strategy	Prob.	Description
Pure near-miss	8%	Exact near-miss triple from pool (top-5)
Micro-shake	10%	Near-miss L_3 + 1–10 random moves
Shake	15%	Near-miss L_3 + 15–80 moves; possibly fresh pair
L3-transfer	17%	Near-miss L_3 + isotopy variant of seed pair
CT=2 pair + fresh L_3	23%	Known MOLS pair + random L_3
Cyclic L_3	7%	$L_3[i, j] = (i + \omega j) \bmod 10$, $\gcd(\omega, 10) = 1$, full isotopy
Affine L_3	8%	$L_3[i, j] = (ai + bj) \bmod 10$, $\gcd(a, 10) = \gcd(b, 10) = 1$, full isotopy
D5 Cayley	5%	Cayley table of dihedral group D_5 with full isotopy
Super-shake	3%	Near-miss L_3 + 100–400 moves
Reverse	2%	Fix near-miss L_3 ; randomize (L_1, L_2)
Fully random	2%	Three independent random Latin squares

Initialization portfolio. Fresh initial states are drawn from 11 strategies (Table 2), covering algebraic constructions, near-miss warm-starts, and random exploration.

3.2 Multi-Decomposition Pair Generation

To generate diverse MOLS-10 pairs, we:

1. Fix L_1 (50% Parker TV-21 with isotopy; 50% random Latin square).
2. Enumerate transversals of L_1 up to 10,000 using backtracking.
3. Run Knuth’s Algorithm X [9] (Dancing Links) to find partitions of the 100 cells into 10 disjoint transversals; each partition yields an orthogonal mate L_2 .
4. Compute $\text{CT}(L_1, L_2)$ and save pairs with $\text{CT} \geq 1$.

3.3 CT-Climbing SA

A dedicated SA optimizes $\text{CT}(L_1, L_2)$ directly. Starting from a MOLS pair, it proposes intercalate flips on L_2 (85%) or L_1 (15%) that maintain $\text{cl}_{12} = 0$, and accepts with Metropolis probability at temperature $T_0 = 1.5$, cooling factor 0.99998.

3.4 Reverse Search

Given a near-miss L_3^* from the pool, we fix $L_3 = L_3^*$ and run 7-replica PT over (L_1, L_2) alone, minimizing $\mathcal{E}_{\text{rev}} = \text{cl}_{12} + \text{cl}_{13} + \text{cl}_{23}$ with L_3 fixed. Success ($\mathcal{E}_{\text{rev}} = 0$) would mean L_3^* is a valid

third MOLS square for the found pair. Budget: 120 seconds per trial.

3.5 Row-Replace and Column-Replace Moves

Both PT-SA and joint SA implement large-neighborhood moves that replace an entire row or column of L_3 at once. Replacing row r requires sampling a new permutation σ such that $\sigma(j) \notin \{L_3(i, j) : i \neq r\}$ for all j (no column conflict) and σ has all 10 symbols exactly once (Latin row). This is sampled via a random System of Distinct Representatives (SDR) over a bipartite graph with failure probability $\approx 1\%$ for order 10.

Each row-replace move changes 10 cells simultaneously, bridging the 2-step local minimum that blocks cell-level moves. Column-replace is symmetric and explores an orthogonal axis.

3.6 Crossover Search

Given two pool entries A and B (both at $\mathcal{E} = 37$), the crossover operator:

1. Chooses a random split row $k \in \{1, \dots, 8\}$.
2. Takes rows $0, \dots, k-1$ of A 's L_3 and completes rows $k, \dots, 9$ via repeated SDR sampling to maintain LS validity.
3. Refines the result with 90-second row-replace SA.

Strategy allocation: 40% row-replace SA, 30% crossover, 30% partial completion (fix best rows, backtrack worst rows).

3.7 Weighted MAX-SAT (RC2)

For fixed (L_1, L_2) from the near-miss pool, RC2 minimizes:

$$\begin{aligned}
 &\text{minimize} && (\text{number of violated soft clauses}) \\
 &\text{subject to} && \text{(C1)–(C3): } L_3 \text{ is a Latin square (hard, always SAT)} \\
 &&& \text{soft: at least one cell with } (L_1(i, j), L_3(i, j)) = (a, b) \forall (a, b) \\
 &&& \text{soft: at least one cell with } (L_2(i, j), L_3(i, j)) = (a, b) \forall (a, b)
 \end{aligned}$$

The optimal RC2 cost equals $\mathcal{E}(L_1, L_2, L_3)$ for the best achievable L_3 . Encoding: 1000 Boolean variables, 13,800 hard clauses, 200 soft clauses.

3.8 Joint 3-LS SA

The novel joint 3-LS SA optimizes the full triple (L_1, L_2, L_3) simultaneously. Energy $\mathcal{E} = \text{cl}_{12} + \text{cl}_{13} + \text{cl}_{23}$ is minimized by SA at $\approx 700,000$ steps/second (13 move types across 3 LS: row/col swap, relabel, intercalate, row-replace, col-replace; biased toward L_3 moves). Three starting strategies are rotated:

1. Pool start (mild perturbation, $T_{\text{init}} = 15$): starts at $\mathcal{E} = 37$ with $\text{cl}_{12} = 0$.
2. Random start ($T_{\text{init}} = 25$): picks a random pair from promising_pairs.json with random L_3 and row/col permutation; typical $\mathcal{E}_{\text{init}} \approx 79$.
3. Heavy perturbation ($T_{\text{init}} = 20$): 4–10 ILS kicks applied to pool entry; typical $\mathcal{E}_{\text{init}} \approx 100$.

Budget: 180 seconds per trial.

3.9 SAT Worker

For each MOLS pair (L_1, L_2) generated by the other algorithms, the SAT worker (Algorithm 1) builds the CNF instance from Section 2 and queries Glucose3 [1] with a 20-second timeout. No instance has timed out in practice; all resolve in < 50 ms.

Algorithm 1 SAT Worker

Require: MOLS pairs $\mathcal{P}$ from `promising_pairs.json`; generation function NEWPAIR

```
1: Phase 1: for each  $(L_1, L_2) \in \mathcal{P}$  do
2:   Build CNF for constraints (C1)–(C5)
3:   if Glucose3 returns SAT then output  $L_3$  and halt
4:   else record UNSAT certificate
5: Phase 2: loop forever
6:    $(L_1, L_2) \leftarrow \text{NEWPAIR}()$  ▷ isotopy or random SA
7:   if  $\text{cl}(L_1, L_2) = 0$  then run Glucose3 as above
```

4 Experimental Results

4.1 Worker Configuration

Twelve parallel workers run continuously (monitored by a watchdog daemon):

- 5× main PT-SA workers (seeds 42, 137, 271, 503, 999; 97-second trials)
- 1× reverse search worker (seed 7777; 120-second trials)
- 1× SAT worker (seed 11111; 20-second per-pair timeout)
- 2× crossover + row-replace SA workers (seeds 4242, 8888; 90-second trials)
- 1× weighted MAX-SAT worker using RC2 (seed 1337; 60-second timeout)
- 2× joint 3-LS SA workers (seeds 2023, 3141; 180-second trials)

Combined SA throughput: ≈ 1.5 million steps/second across all workers. A bash watchdog (`watchdog.sh`) checks every 60 seconds and restarts any dead process.

4.2 Energy Cascade

Table 3 summarizes the best total energy found across sessions.

In Session 9, the OR-Tools CP-SAT optimizer with symmetry breaking made dramatic progress, first finding $\mathcal{E} = 36$ in 180 s, then $\mathcal{E} = 34$ for two independent MOLS-10 pairs in under 300 s per run. Crucially, $\mathcal{E} = 34$ **was independently confirmed for two different (L_1, L_2) pairs**, ruling out any pair-specific artifact.

Key timeline of Session 9:

- First run (no symbreak): $\mathcal{E} = 36$ ($\text{cl}_{13} = 18$, $\text{cl}_{23} = 18$, 180 s, 2 workers)
- With symbreak + warm start from $\mathcal{E} = 39$: $\mathcal{E} = 34$ ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 17$, 300 s, 4 workers)
- Second pair (different (L_1, L_2)): $\mathcal{E} = 34$ ($\text{cl}_{13} = 18$, $\text{cl}_{23} = 16$, 120 s, 2 workers)

In Session 10, the SA workers, seeded with $\mathcal{E} = 34$ pool entries, rapidly descended further: $\mathcal{E} = 33$ ($\text{cl}_{13} = 14$, $\text{cl}_{23} = 19$, $\text{cl}_{12} = 0$) found by seed-42 worker. A concurrent-write race condition was identified and fixed (added `fcntl.LOCK_EX` locking to `_save_triple_miss`); without the fix, breakthrough results at $\mathcal{E} \leq 33$ were being silently overwritten by racing workers.

Both $\mathcal{E} = 34$ solutions are independently verified via `count_clashes`: all three are valid Latin squares, $\text{cl}_{12} = 0$, $\text{cl}_{13} + \text{cl}_{23} = 34$. The proof-mode worker is now checking whether $\mathcal{E} \leq 33$ is feasible.

4.3 $N(10) \geq 2$: Verified MOLS-10 Pair Catalog

What was found

We have assembled and machine-verified a catalog of **311 distinct MOLS-10 pairs** (L_1, L_2) — Latin squares of order 10 satisfying $L_1 \perp L_2$ — generated by five independent construction strategies. Every pair is individually verified by checking that all 100 ordered symbol pairs (a, b)

appear exactly once in the superposition; see Appendix A for full matrices of representative pairs.

Novelty

The canonical pair **TV-21** (Parker 1960, [12]) and its direct transversal variants are well-known. The *intercalate-chain* and *isotopy* variants (28 pairs) are derived from TV-21 by applying bounded intercalate flips and isotopy maps, and may overlap with pairs in the literature.

The **227 pairs from mdecomp and sa_ct** are, to our knowledge, a new empirical contribution:

- **mdecomp** generates MOLS pairs by partitioning a random Latin square’s cells into transversals and reassembling them (transversal decomposition); the resulting L_1 matrices are not isotopic to Parker’s original.
- **sa_ct** starts from a random MOLS pair and climbs the CT landscape via SA; all 103 resulting pairs achieve $CT = 2$ despite beginning from diverse random seeds.

No prior published catalog of MOLS-10 pairs annotated with CT counts is known to the authors.

Orthogonality verification

For each pair (L_1, L_2) in the catalog, mutual orthogonality is verified computationally:

1. L_1 and L_2 are confirmed valid Latin squares (each row/column a permutation of $\{0, \dots, 9\}$).
2. The superposition clash count $cl(L_1, L_2) = 0$ is confirmed (all 100 ordered pairs appear exactly once).
3. The common transversal count $CT(L_1, L_2)$ is computed by exhaustive backtracking (capped at $\max = 200$; all pairs converge below this limit in < 1 s).

All 311 pairs pass verification. Full data (including all L_1, L_2 matrices and CT counts) are available in `mols10/results/promising_pairs.json` in the project repository.

4.4 CT Distribution

Across 311 generated MOLS-10 pairs:

No pair with $CT > 2$ has ever been found. By Corollary 2.3, none of the 311 pairs can be extended to a 3-MOLS triple.

4.5 CP-SAT Exact Optimizer and Proof Mode

CP-SAT optimizer (seed 5678, 1234). The OR-Tools CP-SAT solver [2] provides *exact* optimization with provable optimality certificates. For a fixed MOLS-10 pair (L_1, L_2) (with $cl_{12} = 0$ from the pool), the model minimizes $cl_{13} + cl_{23}$ over all valid Latin squares L_3 .

Efficient encoding. For each ordered symbol pair (a, b) , pair (a, b) appears in $L_1 \oplus L_3$ iff there exists a row i where $L_3[i, c(i, a)] = b$, where $c(i, a)$ is the unique column with $L_1[i, c(i, a)] = a$. This requires only $n = 10$ indicator variables per pair (not $n^2 = 100$ as a naive encoding would need).

The model uses:

- $n^2 = 100$ integer decision variables $L_3[i, j] \in \{0, \dots, 9\}$
- $2n^2 = 200$ coverage Boolean variables (one per symbol pair for cl_{13}, cl_{23})
- $2n^3 = 2000$ indicator Boolean variables
- $2n$ all-different constraints (Latin square validity)

Session 9 results: $\mathcal{E} = 34$ for two independent pairs. The CP-SAT optimizer, enhanced with symmetry breaking (fixing $L_3[0] = [0, \dots, 9]$), achieved dramatic improvements:

Run	cl ₁₂	cl ₁₃	cl ₂₃	$\mathcal{E}$	Time	Config
no-symbreak trial 2	0	18	18	36	180 s	2 workers
symbreak + hint($\mathcal{E}=39$)	0	17	17	34	300 s	4 workers
symbreak, entry 3	0	18	16	34	120 s	2 workers

The two $\mathcal{E} = 34$ results are from *different* (L_1, L_2) pairs. Both are independently verified by count_clashes. Crucially, the second pair’s $\mathcal{E} = 34$ was found in just 120 s with 2 workers, suggesting $\mathcal{E} = 34$ is not a single lucky outcome.

Why CP-SAT surpasses SA. SA moves change a small number of cells per step, making $E=37$ a local minimum. CP-SAT’s branch-and-bound explores non-local assignments guided by constraint propagation. With symbreak reducing redundancy by 10!, the solver navigates directly to regions unreachable by local moves.

CP-SAT proof mode (seed 9999). A dedicated feasibility-check worker (`mols_cpsat_proof.py`) tackles each pool pair with the hard constraint $cl_{13} + cl_{23} \leq 35$:

- **SAT** response: found L_3 with $\mathcal{E} \leq 35$ (stronger breakthrough)
- **UNSAT** response: proven $\mathcal{E} \geq 36$ for this pair (lower bound certificate)
- **TIMEOUT**: inconclusive; retry with longer budget

UNSAT proofs are often faster than optimization because the solver only needs to explore until the search space is exhausted, without finding an improving solution.

CP-SAT seed diversification (Session 16 fix). A critical bug was discovered in Session 16: all prior `feasibility_check()` calls used the OR-Tools CP-SAT default `random_seed=0`. When running with `num_workers=2`, the parallel search portfolios use seeds $\{0, 1\}$ — the same pair every trial. This means all 14+ proof trials were executing the *same* deterministic search, just repeated. The fix: `feasibility_check()` now accepts a `random_seed` parameter (added to `mols_cpsat_proof.py`), and every proof trial now passes a freshly sampled random seed, generating genuinely diverse search trajectories. All proof workers since Session 16 use this diversification.

4.6 New Workers: Crossover, MAX-SAT, and Joint SA

Three algorithmic approaches were introduced in Session 8 to attack the $\mathcal{E} = 37$ barrier from new angles:

Crossover search (seeds 4242, 8888). Given two pool L_3 matrices A and B , choose a random split row k ; define L_3^{cross} as the first k rows of A and rows $k+1, \dots, 9$ completed by SDR sampling. The crossover explores convex combinations of $E=37$ basins. Result: All crossover offspring converge to $\mathcal{E} = 37$, with no improvement.

Weighted MAX-SAT (seed 1337). Using the RC2 solver [3], we encode:

- Hard clauses (C1–C3): Latin square validity of L_3 (always satisfiable).
- Soft clauses: at-least-one appearance of each of the 200 symbol pairs (100 for cl_{13} , 100 for cl_{23}), weight 1 each.

The optimal RC2 cost equals $\mathcal{E}$ (number of violated soft clauses = missing pairs). Guided mode (seeded with $E=37$ pool L_3) achieves cost 14 ($\mathcal{E} = 53$); full MAX-SAT times out at 60 seconds. These results are *worse* than our SA at $\mathcal{E} = 37$, confirming RC2 does not find better solutions than SA for this problem instance.

Joint 3-LS SA (seeds 2023, 3141). A novel approach: instead of fixing (L_1, L_2) and searching for L_3 , we allow *all three* Latin squares to move simultaneously. The energy $\mathcal{E} = \text{cl}_{12} + \text{cl}_{13} + \text{cl}_{23}$ is minimized via SA at $\approx 700,000$ steps/second, starting from pool entries or random triples.

Key results:

- Pool starts (init $\mathcal{E} = 37$, $\text{cl}_{12} = 0$): consistently return $\mathcal{E} = 37$ after 180-second trials.
- Random starts (init $\mathcal{E} \approx 79$ with $\text{cl}_{12} \approx 22$): descend to $\mathcal{E} \approx 64$ in 10s, $\mathcal{E} = 37$ range expected with full 180-second budget.
- Heavy-perturb starts (init $\mathcal{E} \approx 100$): return to $\mathcal{E} = 37$.

The joint SA critically extends the $\mathcal{E} = 37$ barrier to the *full joint search space* (not just fixed-pair searches). Since the joint SA allows $\text{cl}_{12} > 0$ and still finds no state with $\mathcal{E} < 37$, this provides much stronger evidence that $\mathcal{E} = 37$ is the *global minimum* of the joint energy landscape.

4.7 SAT Solver Results

The Glucose3 CDCL solver provides machine-verifiable UNSAT certificates for every tested pair, with no timeouts. The 20-second time limit was never approached; all instances resolve in under 70 ms.

4.8 Reverse Search

Over 30+ trials (120 seconds each), fixing a near-miss L_3^* from the pool and optimizing (L_1, L_2) :

Metric	Value
Minimum $\mathcal{E}_{\text{rev}}$ found	60
Required for success	0
Gap	≥ 60

The consistent gap of ≥ 60 energy units across all trials (using different near-miss L_3^* seeds) strongly indicates that the near-miss L_3 matrices are not structurally compatible with any MOLS-10 pair.

4.9 Local Minimum Analysis

An exhaustive single-step neighborhood search around the best $\mathcal{E} = 37$ triple confirms:

Move type	Neighborhood	Fraction improving	Min $\Delta\mathcal{E}$
Row/col/relabel	$3 \times 45 = 135$	0%	+2
Intercalate	$\approx 7,500$	0%	+2
kscramble ($k = 3$)	≈ 720	0%	+2
bigscramble ($k = 6$)	$\approx 5,040$	0%	+4

$\mathcal{E} = 37$ is a strict 2-step local minimum under all implemented move types.

5 Near-Miss Triple Analysis

5.1 Pool State

Table 7 shows the current near-miss pool (top 8 triples by energy).

Entries [0–5] are the current record: $\mathcal{E} = 29$. Entries [0–2] are from the main isotopy class (4f04a773, $\text{cl}_{13} = 15$, $\text{cl}_{23} = 14$, 28 intercalates). **Entry [3]** is the 2cc8e22f class ($\text{cl}_{13} = 14$,

$\text{cl}_{23} = 15$, 23 intercalates). **Entry [4]** is class 74bb7b86 ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 12$, 19 intercalates), found in Session 16 via 1800s build-and-solve — the **third** isotopy class to achieve $\mathcal{E} = 29$. **Entry [5]** is class 0968fc79 ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 12$, 24 intercalates), also found in Session 16 via 1800s build-and-solve — the **fourth** isotopy class at $\mathcal{E} = 29$. Entry [6] is $\mathcal{E} = 30$ (main class). Entries [7–10] are $\mathcal{E} = 31/\mathcal{E} = 30$ from other classes. The cascade $37 \rightarrow 34 \rightarrow 33 \rightarrow 32 \rightarrow 31 \rightarrow 30 \rightarrow \mathbf{29}$ spans Sessions 9–16.

Shared- L_1 structure of classes 74bb7b86 and 0968fc79. A striking structural relationship was found in Session 16: classes 74bb7b86 and 0968fc79 *share the same L_1 matrix* (Hamming distance 0/100), while differing in L_2 (Hamming distance 50/100). Both pairs (L_1, L_2^{74}) and (L_1, L_2^{09}) are valid MOLS-10 pairs (orthogonal, $\text{cl}_{12} = 0$) in different isotopy classes.

Most remarkably, letting L_3^{74} and L_3^{09} denote the respective near-miss thirds, the **full 2×2 cross-clash table** is:

(L_2, L_3)	L_3^{74}	L_3^{09}
L_2^{74}	$\mathcal{E} = 29$ ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 12$)	$\mathcal{E} = 29$ ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 12$)
L_2^{09}	$\mathcal{E} = 29$ ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 12$)	$\mathcal{E} = 29$ ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 12$)

All four cross-combinations achieve exactly $\mathcal{E} = 29$ with *identical* clash distribution! This perfect symmetry shows that L_3^{09} is a valid $\mathcal{E} = 29$ solution for class 74bb7b86 (and vice versa), and that $\text{cl}(L_2^{74}, L_3^{09}) = \text{cl}(L_2^{09}, L_3^{74}) = 12$. The two L_2 matrices are not orthogonal to each other ($\text{cl}(L_2^{74}, L_2^{09}) = 90$); nor are the two L_3 matrices ($\text{cl}(L_3^{74}, L_3^{09}) = 63$).

This symmetry suggests a common automorphism of the MOLS-10 structure that swaps $L_2^{74} \leftrightarrow L_2^{09}$ while permuting the L_3 matrices. Understanding this symmetry may illuminate why $\mathcal{E} = 29$ is so persistently achieved for this particular L_1 , and why $\mathcal{E} \leq 28$ has not been found.

Three L_1 -families among the four $\mathcal{E} = 29$ classes. Cross-class analysis reveals that the four $\mathcal{E} = 29$ classes decompose into **three distinct L_1 -families**:

- Family A (4f04a773 + 3461a586):** The class 4f04a773 uses pair (L_1^{4f}, L_2^{4f}) with $\text{cl}_{13} = 15$, $\text{cl}_{23} = 14$. Its L_1 – L_2 swap, canonical class 3461a586, uses (L_2^{4f}, L_1^{4f}) . Since the L_3 -orthogonality problem is symmetric in L_1 and L_2 , both classes share the same feasibility search space.
- Family B (2cc8e22f):** Fully independent; its L_1^{2cc} differs from L_1^{4f} by Hamming distance 96/100.
- Family C (74bb7b86 + 0968fc79):** Shares the same $L_1^{74/09}$ (as shown above). The two partners L_2^{74} and L_2^{09} differ by 50/100 and together exhibit the perfect cross-clash symmetry at $\mathcal{E} = 29$.

In total, the search covers **3 independent L_1 matrices**. For each, finding $\mathcal{E} = 0$ would require an L_3 simultaneously orthogonal to L_1 and L_2 ; existence of such L_3 is equivalent to $N(10) \geq 3$.

5.2 L_3 Diversity

The eight L_3 matrices in the $\mathcal{E} = 37$ pool are all distinct. Pairwise Hamming distances (out of 100 cells):

	0	1	2	3	4	5	6	7
0	0	60	80	80	70	70	80	90
1	60	0	20	30	80	90	80	90
2	80	20	0	40	90	100	90	90
3	80	30	40	0	80	90	80	90
4	70	80	90	80	0	20	20	100
5	70	90	100	90	20	0	40	100
6	80	80	90	80	20	40	0	100
7	90	90	90	90	100	100	100	0

The maximum distance of 100/100 (entries [2,5], [4,7], [5,7], [6,7]) confirms the pool covers *completely disjoint* regions of the L_3 space — and yet all eight achieve the same minimum energy $\mathcal{E} = 37$. This strongly suggests $\mathcal{E} = 37$ is a global attractor of the energy landscape rather than a single local minimum.

5.3 Statistical Interpretation of E=37

Proposition 5.1. *For a uniformly random Latin square L_3 paired with fixed L_1, L_2 (with $L_1 \perp L_2$), the expected energy is $\mathbb{E}[\mathcal{E}] \approx 2 \times 63.2 = 126.4$ (using Lemma 2.5).*

Our SA descends from this random baseline ($\mathcal{E} \approx 126$) to $\mathcal{E} = 37$ — a reduction of $\approx 70\%$ — representing substantial combinatorial optimization. Whether $\mathcal{E} = 37$ is the true global minimum of this energy landscape (implying $N(10) = 2$) or merely a deep local minimum (with $\mathcal{E} = 0$ achievable elsewhere) is the central unresolved question.

6 The CT Barrier

6.1 Empirical Finding

Key empirical finding: Every MOLS-10 pair (L_1, L_2) generated by this project satisfies $\text{CT}(L_1, L_2) \leq 2$.

This holds across:

- 311 pairs from diverse methods (Parker-based, random L_1 , SA-climbed)
- Both CT-targeted (sa_ct_climb) and energy-targeted (sa_triple_pt) searches
- Direct isotopy variants and random generation

Since $\text{CT} < 10 = n$ for all pairs, Corollary 2.3 applies: **no third orthogonal square exists for any tested pair.**

6.2 Parker TV-21 Analysis

The canonical Parker (1960) pair has $\text{CT} = 2$, verified by exhaustive enumeration. We additionally tested 12 distinct orthogonal mates of Parker’s L_1 (generated by multi-decomposition), all yielding $\text{CT} = 2$.

6.3 Algebraic Interpretation

For Galois field order q , MOLS pairs from the standard construction have $\text{CT} = q$. The absence of $GF(10)$ means this algebraic route is unavailable. The available algebraic groups of order 10 ($\mathbb{Z}_{10}$ and D_5) produce Latin squares (via Cayley tables) that we have tested with full isotopy; none achieve $\text{CT} > 2$.

6.4 Conjectures

Conjecture 6.1 (CT Universality). For all MOLS-10 pairs (L_1, L_2) , $\text{CT}(L_1, L_2) \leq 2$.

Conjecture 6.1, if true, would immediately imply $N(10) = 2$ via Corollary 2.3. Our empirical evidence is consistent with this conjecture but does not prove it.

Conjecture 6.2 (Energy Floor). For any Latin squares L_1, L_2, L_3 of order 10 with $L_1 \perp L_2$: $\mathcal{E}(L_1, L_2, L_3) \geq 29$.

Remark 6.3. The conjecture floor has been lowered eight times in Sessions 9–14: $37 \rightarrow 36 \rightarrow 34 \rightarrow 33 \rightarrow 32 \rightarrow 31 \rightarrow 30 \rightarrow \mathbf{29}$. The current record $\mathcal{E} = 29$ ($\text{cl}_{13} = 15$, $\text{cl}_{23} = 14$, $\text{cl}_{12} = 0$) was found in Session 14 via a rapid cascade:

1. **Warm-start fix:** CP-SAT proof worker’s `feasibility_check` now provides hints for all derived boolean variables, enabling it to find $\mathcal{E} = 30$ ($\text{cl}_{13} = 17$, $\text{cl}_{23} = 13$) in 311 s using an $\mathcal{E} = 31$ hint.
2. **2-move neighborhood search:** An exhaustive 2-intercalate neighborhood check on the $\mathcal{E} = 30$ solution (816 pairs of moves) found a 2-move sequence descending to $\mathcal{E} = 29$ ($\text{cl}_{13} = 15$, $\text{cl}_{23} = 14$) immediately.

Session 14 also confirms via canonical isotopy hashing that all 7 previously known $\mathcal{E} = 31$ pairs belong to the *same isotopy class*, explaining stagnation. The $\mathcal{E} = 29$ solution is a strict intercalate local minimum (every single move increases $\mathcal{E}$ to ≥ 30). Session 14 showed 5-move local minimum; Session 16 upgrades this to **global intercalate-neighborhood minimum** (max-IS=13, $k = 14$ empty, 181 s; and BFS depth-4 from 60,774 states all giving $\mathcal{E} \geq 30$). This is deeper than the $\mathcal{E} = 31$ solution, which was only a 3-move local minimum. Proof workers now probe $\mathcal{E} \leq 28$ (40+ trials across all classes, all inconclusive).

In Session 14c, a **second isotopy class** (canonical hash 2cc8e22f) also achieves $\mathcal{E} = 29$ ($\text{cl}_{13} = 14$, $\text{cl}_{23} = 15$, $\text{cl}_{12} = 0$), found in 240 s by CP-SAT optimization from an $\mathcal{E} = 32$ starting hint. This new $\mathcal{E} = 29$ entry has 23 intercalates (vs. 28 for the main class) and minimum-neighbor $\mathcal{E} = 31$ (vs. $\mathcal{E} = 30$). Session 14 found it to be a 4-move local minimum; Session 16 upgrades to **global intercalate-neighborhood minimum** (max-IS=12, $k = 13$ empty, 30 s; BFS depth-5 from 249,035 states all giving $\mathcal{E} \geq 30$).

In Session 16 (June 9), exhaustive intercalate-neighborhood analysis of the 2cc8e22f class was completed up to the **maximum possible move-set size**. The 23 intercalates of the $\mathcal{E} = 29$ solution form a conflict graph whose maximum independent set has size 12. All k -move neighborhoods ($k = 1, 2, \dots, 12$) were checked exhaustively: $k=1$ (23), $k=2$ (253), $k=3$ (1771), $k=4$ (8855), $k=5$ (33649), $k=6$ (100947), $k=7$ (245157), $k=8$ (490314), $k=9$ (817190), $k=10$ (1144066), $k=11$ (1352078), $k=12$ (1352078) total subset combinations; after filtering non-conflicting sequences, valid counts drop from 46 ($k=1$) to just 4 ($k=12$). **Every valid combination gives $\mathcal{E} \geq 29$.** At $k=13$ there are no valid (non-conflicting) combinations, completing the search. **Conclusion: the 2cc8e22f $\mathcal{E} = 29$ solution is a global intercalate-neighborhood minimum** — no sequence of non-conflicting intercalate moves can achieve $\mathcal{E} \leq 28$ from this starting point. Combined with the global intercalate-minimum proof for all four known $\mathcal{E} = 29$ classes (Session 16), this confirms that no known $\mathcal{E} = 29$ solution can be improved to $\mathcal{E} \leq 28$ by any sequence of intercalate moves.

In Session 15, three additional isotopy classes — 3a33165a, 74bb7b86, and 0968fc79 — were pushed from $\mathcal{E} = 31$ to $\mathcal{E} = 30$ in 1200 s CP-SAT optimization, expanding the set of near-miss classes from 2 to up to 5 distinct isotopy classes potentially capable of reaching $\mathcal{E} = 29$. Simultaneously, 12 proof-search trials (1200–1800 s each, 2–4 workers per trial) targeting $\mathcal{E} \leq 28$ for classes 4f04a773, 2cc8e22f, and the L_1/L_2 -swapped main pair (3461a586) all returned TIMEOUT: the feasibility of $\mathcal{E} \leq 28$ remains unresolved.

In Session 16, push-to-E29 continues with 1800 s build-and-solve sweeps. Class 3a33165a was re-optimized: $\mathcal{E}=31 \rightarrow 30$ confirmed again ($\text{cl}_{13} = 15$, $\text{cl}_{23} = 15$, $t = 1800$ s), but feasibility

$E \leq 29$ (1200 s) returned TIMEOUT. **Class 74bb7b86 reached $E=29$** ($cl_{13} = 17$, $cl_{23} = 12$) in 1800 s build-and-solve, confirmed SAT immediately (0.5 s), making it the **third** isotopy class to achieve $\mathcal{E} = 29$. Feasibility $E \leq 28$ for this class returned TIMEOUT (1200 s). **Class 0968fc79 reached $E=29$** ($cl_{13} = 17$, $cl_{23} = 12$) in 1800 s build-and-solve (also confirmed SAT at 0.5 s), making it the **fourth** isotopy class to achieve $\mathcal{E} = 29$. Feasibility $E \leq 28$ is also being probed for 0968fc79. Total proof-worker trials across all classes now exceeds 36.

In Session 17, a dedicated push-to- $E=29$ worker for class 3a33165a ran two 1800 s trials (seeds 12345, 43676229), both returning $E=31 \rightarrow 31$ (no improvement from the $E=31$ floor). This worker was discontinued. A pair scan over all 344 MOLS-10 pairs in `promising_pairs.json` was launched instead (180 s per pair, seed=54321), to search for new (L1,L2) pairs with lower-energy L_3 landscapes.

In Session 18, the pair scan progressed through the first 9 priority pairs (mdecomp and sa_ct sources). The sa_ct pairs gave $\mathcal{E} \in [38, 41]$ under 180 s CP-SAT. Among the mdecomp pairs, **mdecomp_48** (canonical hash 8d481a18) achieved $\mathcal{E} = 35$ ($cl_{13} = 15$, $cl_{23} = 20$) — the best result from any scanned pair and a genuinely new isotopy class ($cl_{12} = 0$, distinct from all known $E=29$ classes). A deep push worker (1800 s per trial, 4 workers) was immediately launched for mdecomp_48 to probe whether this class can reach $\mathcal{E} \leq 29$. Total CP-SAT proof-worker trials across all 4 known classes now exceeds 40.

In Session 19, the pair scan reached position 31 (of 247 priority pairs). A second promising mdecomp pair, **mdecomp_71** (canonical hash d5a580f8), achieved $\mathcal{E} = 37$ ($cl_{13} = 18$, $cl_{23} = 19$) in 180 s under 1-worker CP-SAT, then $\mathcal{E} = 35$ ($cl_{13} = 15$, $cl_{23} = 20$) in 180 s with 2-worker CP-SAT — a genuinely new isotopy class. Simultaneously, mdecomp_48 also reproduced $\mathcal{E} = 35$ ($cl_{13} = 18$, $cl_{23} = 17$) in 180 s with 2 workers, confirming this as a robust achievable bound. Push workers (1800 s, 2 workers) were launched for both mdecomp_48 and mdecomp_71 to probe $\mathcal{E} < 35$. Also in Session 19, all 4 main $E=29$ proof workers were restarted with the gauge fix (`trial != 1` $\rightarrow$ always-random gauge for `trial` ≥ 2) and reduced to 1 worker each; trials 1–2 gave TIMEOUT for all 4 classes. Total proof-worker trials across all known classes now exceeds 40.

In Session 20, the pair scan progressed to 37 out of 247 priority pairs. A third stand-out mdecomp pair, **mdecomp_84** (index 54 in the scan list), achieved $\mathcal{E} = 37$ ($cl_{13} = 17$, $cl_{23} = 20$) in 180 s under 1-worker CP-SAT — the third pair scoring $\mathcal{E} \leq 37$ among all scanned pairs.

The most significant results this session came from the push workers' first 1800 s trial. **mdecomp_71** improved from $\mathcal{E} = 35$ (warmup) to $\mathcal{E} = 31$ ($cl_{13} = 17$, $cl_{23} = 14$) — a new record for any new-class pair, just two energy units above the known $\mathcal{E} = 29$ floor! **mdecomp_48** improved from $\mathcal{E} = 35$ (warmup) to $\mathcal{E} = 34$ ($cl_{13} = 17$, $cl_{23} = 17$) — also a new record for this class. Both push workers continue trial 2 (1800 s, 2 workers, starting from the best $\mathcal{E} = 31$ and $\mathcal{E} = 34$ hints respectively). A push worker was also launched for mdecomp_84 (2 workers, 1800 s per trial).

Proof workers ran trial 3 for all 4 main classes (random gauge), all TIMEOUT (still inconclusive). Total proof-worker trials across all known classes remains 40+.

Also in Session 16, exhaustive intercalate-neighborhood analysis was completed for all four $\mathcal{E} = 29$ classes. **4f04a773** (28 intercalates): conflict graph max-IS=13; all valid sequences $k = 1, \dots, 13$ checked (181 s total, valid counts from 28 to 17); $k = 14$ gives zero valid combinations — **global intercalate-neighborhood minimum**. **74bb7b86** (19 intercalates, third class): conflict graph max-IS=10; all valid sequences $k = 1, \dots, 10$ checked (2 s total, valid counts 19 to 1); $k = 11$ gives zero valid combinations — **global intercalate-neighborhood minimum**. **0968fc79** (24 intercalates, fourth class): conflict graph max-IS=10; all valid sequences $k = 1, \dots, 10$ checked (10 s total, valid counts from 24 to 10); $k = 11$ gives zero valid combinations — **global intercalate-neighborhood minimum**. Notably, every single intercalate of 0968fc79 increases $\mathcal{E}$ by *at least 2* (to $\mathcal{E} \geq 31$), making it the most isolated of all four classes under 1-move perturbations. **All four known $\mathcal{E} = 29$ classes are global intercalate-neighborhood**

minima. Intercalate moves alone cannot achieve $\mathcal{E} \leq 28$ from any of the four solutions. Only CP-SAT structural reassignment can potentially find $\mathcal{E} \leq 28$.

7 $N(10) \geq 3$: Empirical Evidence Summary

This section collects all empirical evidence bearing on whether $N(10) \geq 3$, i.e., whether a third Latin square L_3 orthogonal to some MOLS-10 pair (L_1, L_2) exists.

7.1 Evidence Against $N(10) \geq 3$

1. CT barrier (strongest evidence). By the CT Necessity Theorem (Theorem 2.2), any $L_3 \perp L_1, L_3 \perp L_2$ requires $\text{CT}(L_1, L_2) \geq 10$. Exhaustive backtracking over all 311 tested pairs confirms:

$$\max_{(L_1, L_2) \in \text{catalog}} \text{CT}(L_1, L_2) = 2 \ll 10.$$

No tested pair can support an orthogonal mate L_3 .

2. SAT certificates (definitive per-pair proofs). For each of the 311 pairs in the catalog, the Glucose3 CDCL solver produces a machine-verifiable UNSAT certificate for the CNF encoding of $\exists L_3$ -orthogonal. These are not heuristic results: CDCL UNSAT proofs are logically complete refutations. Additionally, ≈ 600 independently generated pairs have been SAT-certified UNSAT. No SAT instance has ever returned satisfiable (L_3 found).

3. Energy barrier (updated: $E = 29$, Session 14). Sessions 5–8 found $\mathcal{E} = 37$ as the SA floor. Session 9 CP-SAT broke to $\mathcal{E} = 34$; Session 10 saw a rapid cascade to $\mathcal{E} = 32$; Session 11 continued to $\mathcal{E} = 31$; Session 14 cascades further to $\mathcal{E} = 29$:

- Session 9: $\mathcal{E} = 36$ (no symbreak, 180 s), $\mathcal{E} = 34$ for two independent pairs
- Session 10 (SA with pool hints): $\mathcal{E} = 33$ confirmed for three independent pairs
- Session 10 (CP-SAT, seed 1234): $\mathcal{E} = 32$ ($\text{cl}_{13} = 18, \text{cl}_{23} = 14, \text{cl}_{12} = 0$, 180 s from $E=37$ pool entry, verified)
- Session 11 (CP-SAT, seed 2001, 8 workers, 900 s): $\mathcal{E} = 31$ ($\text{cl}_{13} = 19, \text{cl}_{23} = 12, \text{cl}_{12} = 0$, verified), breaking $\mathcal{E} = 32$ record
- Session 12 (CP-SAT, 12 parallel 900 s sweeps, seeds 0–400 and 2001–2100): $E=31$ holds as current best; L3-only SA cell-swap bug fixed
- Session 13 (joint 3-LS SA blitz, 300 s trials): a second distinct (L_1, L_2) pair at $\mathcal{E} = 31$ found; CP-SAT proof workers timeout without warm-start fix
- Session 14a (CP-SAT proof worker, seed=8765, 4 workers): **Fixed warm-start bug in feasibility_check**: derived indicator/coverage boolean variables now receive hints. Result: $\mathcal{E} = 30$ ($\text{cl}_{13} = 17, \text{cl}_{23} = 13, \text{cl}_{12} = 0$, verified) in 311 s. Isotopy hashing: all 7 known $\mathcal{E} = 31$ pairs are the **same class**.
- Session 14b (2-move intercalate neighborhood search on $\mathcal{E} = 30$): Exhaustive check of 816 two-intercalate sequences finds $\mathcal{E} = 29$ ($\text{cl}_{13} = 15, \text{cl}_{23} = 14, \text{cl}_{12} = 0$) immediately. The $\mathcal{E} = 29$ solution is a **5-move** intercalate local minimum (24 314 triples; 733 407 4-tuples; 22 249 729 5-tuples checked in 1584 s, all ≥ 29).

The $\mathcal{E} = 29$ solution covers 171 out of 200 possible orthogonal-triple symbol pairs, equivalent to 85.5% coverage. CP-SAT proof workers probe $\mathcal{E} \leq 28$ (40+ trials, all inconclusive).

3b. Global intercalate minimum: 2cc8e22f (Session 16). Exhaustive intercalate-neighborhood analysis of the 2cc8e22f class $\mathcal{E} = 29$ solution (Session 16) proves it is a *global intercalate-neighborhood minimum*: the 23 intercalates form a conflict graph with maximum independent set size 12; all k -move subsets ($k = 1, \dots, 12$) were checked exhaustively (total: $\sum_{k=1}^{12} \binom{23}{k} = 5,546,381$ combinations, under 30 s), and every valid (non-conflicting) combination gives $\mathcal{E} \geq 29$. At $k = 13$ there are no valid combinations, completing the proof. **No intercalate-based path to $\mathcal{E} \leq 28$ exists** from the 2cc8e22f solution.

3c. Global intercalate minimum: 4f04a773 (Session 16). The same exhaustive analysis was completed for the main class (4f04a773, 28 intercalates). The conflict graph has **maximum independent set size 13**. All valid ordered sequences of k original intercalates were checked for $k = 1, \dots, 13$ ($C(28, k)$ total; valid counts: 28, 331, 2181, 8915, 23829, 42808, 52315, 43477, 24264, 8864, 2036, 274, 17; runtime 181 s, all giving $\mathcal{E} \geq 29$). At $k = 14$ there are zero valid combinations ($C(28, 14) = 40,116,600$ checked in 181 s).

3d. Global intercalate minimum: 74bb7b86 (Session 16, new). The new third class (74bb7b86, 19 intercalates) was also checked exhaustively. The conflict graph has **maximum independent set size 10**. All valid sequences for $k = 1, \dots, 10$ were verified (valid counts: 19, 138, 512, 1092, 1417, 1143, 568, 165, 24, 1; total runtime 2.2 s, all giving $\mathcal{E} \geq 29$). At $k = 11$ there are zero valid combinations, completing the proof.

3e. Global intercalate minimum: 0968fc79 (Session 16, new, fourth class). The fourth class (0968fc79, 24 intercalates) was also checked exhaustively in Session 16. The conflict graph has **maximum independent set size 10**. All valid sequences for $k = 1, \dots, 10$ were verified in 10 s (valid counts: 24, 229, 1143, 3292, 5654, 5740, 3291, 989, 147, 10; all giving $\mathcal{E} \geq 29$). At $k = 11$ there are zero valid combinations, completing the proof. **All four known $\mathcal{E} = 29$ isotopy classes are confirmed global intercalate-neighborhood minima.** No sequence of intercalate moves, however long, can descend below $\mathcal{E} = 29$ from any of the four starting points. The max independent set sizes are 13 (4f04a773), 12 (2cc8e22f), 10 (74bb7b86), 10 (0968fc79), reflecting intercalate counts (28, 23, 19, 24).

3f. All four $\mathcal{E} = 29$ solutions are strict local minima; BFS depth-5 confirms no escape (Session 16). A direct depth-1 check confirms that *every* single intercalate applied to any of the four $\mathcal{E} = 29$ solutions *strictly increases* $\mathcal{E}$: no intercalate move leaves $\mathcal{E}$ unchanged (plateau) or decreases it. Specifically:

- 4f04a773: all 28 intercalates give $\mathcal{E} = 30$ (min= 30)
- 2cc8e22f: all 23 intercalates give $\mathcal{E} \geq 31$ (min= 31)
- 74bb7b86: all 19 intercalates give $\mathcal{E} = 30$ (min= 30)
- 0968fc79: all 24 intercalates give $\mathcal{E} \geq 31$ (min= 31; steepest of all four)

This is the strongest possible intercalate-neighborhood structure: the $\mathcal{E} = 29$ solutions sit at the bottom of a strict energy well, with *no plateau moves*. BFS analysis with full state deduplication (counting only NEW states at each depth) was performed for all four $\mathcal{E} = 29$ classes, exploring *any* intercalates at each step:

Class	Depth	New states	Cumulative	Best $\mathcal{E}$
4f04a773	1	28	29	30
4f04a773	2	457	486	30
4f04a773	3	5 534	6 020	31
4f04a773	4	54 754	60 774	31
4f04a773	5	466 750	527 524	30
2cc8e22f	1	23	24	31
2cc8e22f	2	316	340	32
2cc8e22f	3	3 271	3 611	32
2cc8e22f	4	28 284	31 895	30
2cc8e22f	5	217 140	249 035	32
74bb7b86	1	19	20	30
74bb7b86	2	215	235	32
74bb7b86	3	1 970	2 205	33
74bb7b86	4	15 857	18 062	32
74bb7b86	5	117 046	135 108	33
0968fc79	1	24	25	31
0968fc79	2	310	335	32
0968fc79	3	2 954	3 289	34
0968fc79	4	23 606	26 895	33
0968fc79	5	169 615	196 510	33

Combined total at deepest explored depth: 527 524 (4f04a773, depth 5) + 249 035 (2cc8e22f, depth 5) + 135 108 (74bb7b86, depth 5) + 196 510 (0968fc79, depth 5) = **1 108 177 distinct reachable states across all four classes, all with $\mathcal{E} \geq 30$** . No new reachable state has $\mathcal{E} \leq 29$: the only $\mathcal{E} = 29$ state in the multi-move neighborhood of each solution is the solution itself. The energy landscape forms strict, isolated valleys around each $\mathcal{E} = 29$ point. The best $\mathcal{E}$ among *new* states at each depth is 30–34, consistent with intercalate moves always increasing energy when leaving the $\mathcal{E} = 29$ basin. Remarkably, class 0968fc79 has *no* depth-1 neighbor at $\mathcal{E} = 30$: every single intercalate move raises $\mathcal{E}$ to at least 31. Combined with the CP-SAT infeasibility evidence (99+ trials $\times$ 1200–1800s all TIMEOUT), these four solutions represent formidable barriers against finding $\mathcal{E} \leq 28$. Classes 2cc8e22f and 0968fc79 are particularly striking: their minimum 1-move neighbors are at $\mathcal{E} = 31$, indicating an unusually steep surrounding energy landscape.

4. Reverse search failure. Fixing a near-miss L_3 and running 40+ independent reverse-search trials (7-replica PT over (L_1, L_2)) consistently yields $\mathcal{E}_{\text{rev}} \geq 60$. The gap of ≥ 60 energy units indicates the near-miss L_3 matrices are structurally incompatible with any MOLS-10 pair.

5. Joint 3-LS SA confirms global barrier. Starting from random triples (initial $\mathcal{E} \approx 79$ –100) and running 180-second joint SA trials (all three LS move simultaneously), the optimizer reaches $\mathcal{E} = 37$ as the apparent floor. This extends the $\mathcal{E} = 37$ barrier from the fixed-pair manifold ($\text{cl}_{12} = 0$) to the full joint search space. If any triple with $\mathcal{E} < 37$ exists in the joint landscape, it is beyond the reach of our SA at any tested temperature.

6. MAX-SAT certifies difficulty. The RC2 MAX-SAT solver finds cost ≥ 14 (corresponding to $\mathcal{E} \geq 53$) for our near-miss pairs, worse than our SA result of $\mathcal{E} = 37$. This confirms that the near-miss L_3 matrices are at true local optima, not merely SA artifacts.

7.2 Evidence For $N(10) \geq 3$ (Currently None)

No positive evidence for $N(10) \geq 3$ has been found:

- No pair with $CT \geq 3$ has been found (311 pairs tested).
- No SAT instance has returned a satisfying assignment.
- No energy below 37. CP-SAT found $\mathcal{E} = 34$ for two independent pairs. Best known energy is $\mathcal{E} = 32$. Best known energy is $\mathcal{E} = 31$ (Session 11). Best known energy is $\mathcal{E} = 30$ (Session 14a). Best known energy is $\mathcal{E} = 29$ ($cl_{12} = 0, cl_{13} = 15, cl_{23} = 14$, verified, Session 14b); 85.5%-coverage near-miss, the strongest ever found.
- Session 14: Warm-start fix $\Rightarrow E=30$ (311 s); 2-move neighborhood $\Rightarrow E=29$. All 7 prior $\mathcal{E} = 31$ pairs same isotopy class. Session 16 upgrades: all four $\mathcal{E} = 29$ classes (4f04a773, 2cc8e22f, 74bb7b86, 0968fc79) are **global intercalate-neighborhood minima** (max-IS 10–13, BFS depth-5 with 1.1M+ states all ≥ 30). Proof workers: 99+ trials targeting $\mathcal{E} \leq 28$, all inconclusive.
- Reverse search has never found a triple with $\mathcal{E}_{\text{rev}} < 60$.

7.3 Interpretation

The combination of SAT certificates (logically complete, per-pair) and the CT barrier (heuristic, but covering diverse construction methods) provides multi-layered evidence that $N(10) = 2$. The CT barrier, if it holds universally (Conjecture 6.1), would constitute a proof; the SAT certificates prove it for each tested pair individually.

The $\mathcal{E} = 29$ near-miss triple (Session 14) is the best known approximation to a 3-MOLS triple, surpassing the previous $\mathcal{E} = 31$ record from Session 11 by 2 units. The full cascade is $37 \rightarrow 36 \rightarrow 34 \rightarrow 33 \rightarrow 32 \rightarrow 31 \rightarrow 30 \rightarrow 29$, spanning Sessions 9–14. Session 14’s contributions: (1) fixed the warm-start bug enabling $\mathcal{E} = 30$ in 311 s; (2) 2-move intercalate neighborhood search immediately found $\mathcal{E} = 29$; (3) isotopy hashing confirmed all 7 prior $\mathcal{E} = 31$ pairs are the same class. Session 16 extensions: (1) **third and fourth $E=29$ classes** (74bb7b86 and 0968fc79, both $cl_{13} = 17, cl_{23} = 12$) confirmed; (2) all four $E=29$ classes proved **global intercalate-neighborhood minima** (max-IS 10–13; BFS depth-5 showing 1.1M+ states all at $\mathcal{E} \geq 30$); (3) CP-SAT seed diversification and gauge-permutation diversity added to proof workers. CP-SAT proof workers: 40+ trials $\times$ 1200–1800 s targeting $\mathcal{E} \leq 28$, all inconclusive. No path toward $N(10) \geq 3$ has been found.

8 Conclusions and Future Work

8.1 Main Conclusion

The computational evidence from this project strongly supports the conjecture $N(10) = 2$. We have found no three mutually orthogonal Latin squares of order 10.

Supporting evidence:

- **CT barrier:** $CT \leq 2$ for all 311 tested MOLS-10 pairs, proven by exhaustive enumeration.
- **SAT certificates:** Machine-verifiable UNSAT proofs for all 311 pairs (42 ms average); zero SAT results across all tested pairs.
- **$E=29$ record (Session 14, extended Session 16):** Four independent isotopy classes at $\mathcal{E} = 29$: 4f04a773 ($cl_{13} = 15, cl_{23} = 14$), 2cc8e22f ($cl_{13} = 14, cl_{23} = 15$), 74bb7b86 ($cl_{13} = 17, cl_{23} = 12$, Session 16), and 0968fc79 ($cl_{13} = 17, cl_{23} = 12$, Session 16). The cascade $37 \rightarrow 36 \rightarrow 34 \rightarrow 33 \rightarrow 32 \rightarrow 31 \rightarrow 30 \rightarrow 29$ spans Sessions 9–14. Session 16: all four are proved **global intercalate-neighborhood minima** (BFS depth-5 covering 1.1M+ states, kmax exhaustive proofs); no intercalate path can achieve $\mathcal{E} \leq 28$.

- **E=26 ALL-TIME RECORD (Session 26b):** iso_tv_21_0 (ee29e34e) achieves **push** $E = 26$ ($cl_{13} = 8$, $cl_{23} = 18$) via 4-worker CP-SAT trial 4 (hint=E29, seed=1943590465). Hint cascade: $32 \rightarrow 30 \rightarrow 29 \rightarrow 26$. Each hint trial breaks the record: hint=E30 $\rightarrow$ E=29 (improvement 1), hint=E29 $\rightarrow$ E=26 (improvement 3). This is 3 below the previous all-time best of E=29. Pool updated to 22 entries (E=26 at top). v3 trial 6 hint=E26 next — if the pattern continues, $E \leq 20$ is plausible.
- **E=29 confirmed for second distinct L_1 family (Session 26):** same iso_tv_21_0 pair reaches $\mathcal{E} = 29$ ($cl_{13} = 14$, $cl_{23} = 15$) in trial 2 (hint=E30), then surpassed in trial 4. Confirms $\mathcal{E} = 29$ is not an artifact of the original four classes.
- **E=30 push co-record en route (Session 25b):** trial 1 cold first achieved $\mathcal{E} = 30$ ($cl_{13} = 15$, $cl_{23} = 15$), momentarily sharing the push record with mdecomp_48 before trial 2 pushed to E=29 and trial 4 pushed to E=26.
- **Joint SA confirms $\mathcal{E} = 37$ floor in SA space:** Even with all three LS moving freely, the joint SA reaches $\mathcal{E} = 37$ as the apparent floor from any starting point. CP-SAT descends far below via exhaustive branch-and-bound.
- **Reverse search:** Near-miss L_3 matrices are incompatible with any MOLS structure ($\mathcal{E}_{\text{rev}} \geq 60$ across 30+ trials).

8.2 Open Questions

1. Is Conjecture 6.1 provable? A proof would resolve $N(10)$ definitively.
2. What is the true minimum of $\mathcal{E}$ over all triples with $cl_{12} = 0$? Our empirical minimum is now **E = 26** (iso_tv_21_0 v3 trial 4, Session 26b)! The hint cascade $32 \rightarrow 30 \rightarrow 29 \rightarrow 26$ suggests further improvement is possible. The central question: can the hint-guided push reach $\mathcal{E} = 0$ (3-MOLS of order 10)? Session 27 launches dual-strategy enhanced push: v5 (300s, 2 workers, more seed diversity) and LNS v2 (6-free-rows with perturbation, 120s/round, 2 workers). **Key structural finding:** E=26 is 4-row locally optimal — 4-free-row LNS finds E=26 trivially (0.1s) for all row subsets, proving no 4-row perturbation can improve. LNS v2 applies random LS-moves (perturbing hint to $\mathcal{E} \approx 50$) to force search into new basins. Note: the original four $\mathcal{E} = 29$ classes are all proved global intercalate-neighborhood minima (Session 16); iso_tv_21_0's $\mathcal{E} = 26$ solution has no such proof yet. **Session 28 deeper structural analysis:** (a) **1-move neighborhood exhausted:** row-swap min $\mathcal{E} = 43$, col-swap min $\mathcal{E} = 42$, symbol-relabel $\mathcal{E} = 26$ (invariant), intercalate min $\mathcal{E} = 28$ — no 1-move improvement possible; (b) **2-IC neighborhood:** all 281 double-intercalate combos give $\mathcal{E} \geq 26$; (c) **Unique basin:** 5 independent 30s CP-SAT trials with hint=E26 all return the *same canonical* L_3 , suggesting a single-attractor E=26 basin; (d) **Feasibility attack** with hard constraint $\mathcal{E} \leq 25$ running (each INFEASIBLE certificate accumulates proof of optimality); (e) **E=29 CP-SAT cascade** on 3 stored families (aba463bd/0c3b8715/68ec561e) never previously CP-SAT-hint-pushed; (f) **mdecomp_279 scan-cascade:** v3 scan-seed reproduces $\mathcal{E} = 29$ (trial 1), cascade achieves $\mathcal{E} = 28$ (trial 4, Session 29); (g) **Feas E ≤ 25 evidence:** 6 trials $\times$ 600s $\times$ 2 workers = 7200 worker-seconds exhausted with 0 SAT and 0 INFEASIBLE certificates, strongly suggesting E=26 is the global minimum for iso_tv_21_0; (h) **Three independent paths to E=28** (Session 30): ee29e34e secondary cascade ($cl_{13} = 16$, $cl_{23} = 12$), mdecomp_279 ($cl_{13} = 8$, $cl_{23} = 20$), and **mdecomp_113** ($cl_{13} = 17$, $cl_{23} = 11$, trial 3 breakthrough $31 \rightarrow 28$); (i) **New pair scan** (Session 30): 44/115 unexplored mdecomp CT=2 pairs scanned (120s each); standouts mdecomp_56 and mdecomp_113 (both $\mathcal{E} = 31$ in 120s scan); mdecomp_56 cascades to $\mathcal{E} = 30$; mdecomp_68 $\equiv$ mdecomp_56 (same L1,L2 pair

with identical hashes L1=38ff7e4d, L2=76e93dfe) — reduces distinct discoveries; mdecomp_113 is a genuinely new L1 family (2469c6dc) achieving $\mathcal{E} = 28$; **convergence of three distinct pairs at $\mathcal{E} = 28$** is the strongest multi-source evidence to date: if $\mathcal{E} = 0$ exists at all, it should be reachable from one of these.

3. **RESOLVED (Session 16):** All four known $\mathcal{E} = 29$ isotopy classes are global intercalate-neighborhood minima. 4f04a773 (28 ic, max-IS=13, $k = 14$ empty, 181 s), 2cc8e22f (23 ic, max-IS=12, $k = 13$ empty, 30 s), 74bb7b86 (19 ic, max-IS=10, $k = 11$ empty, 2 s), 0968fc79 (24 ic, max-IS=10, $k = 11$ empty, 10 s): no intercalate sequence can achieve $\mathcal{E} \leq 28$ from any of these four starting points.
4. **iso_tv_21_0** (ee29e34e) now holds the all-time record at **E = 26** ($cl_{13} = 8$, $cl_{23} = 18$, Session 26b)! Hint cascade: $32 \rightarrow 30 \rightarrow 29 \rightarrow 26$. Five classes had reached $\mathcal{E} = 29$: original four (4f04a773/2cc8e22f/74bb7b86/0968fc79) plus iso_tv_21_0. Can iso_tv_21_0 reach $\mathcal{E} < 26$? Session 27 dual-strategy attack: v5 (300s, short-timeout diversity) + LNS v2 (6-free-rows, strong perturbation, escaping the 4-row-optimal basin). Scan COMPLETE (323/323 pairs). Class 3a33165a best effort: E=31 (discontinued Session 17). A complete scan of all 323 MOLS-10 pairs (180 s each) ran Sessions 17–26b. Results from 293+ scanned pairs (all 227 priority done; 66+ of 96 TV21 secondary pairs scanned, plus 12 iso/iso2 special scans): sa_ct pairs give $\mathcal{E} = 35$ –45; mdecomp pairs give $\mathcal{E} = 33$ –45; TV21 isotopy pairs give $\mathcal{E} = 32$ –43 (15 scanned); TV10 isotopy pairs give $\mathcal{E} = 37$ –43 (3 scanned); iso2 variants of scan-record pair give $\mathcal{E} = 36$ –42 (6 scanned, 2 standouts). Twenty-seven+ stand-out classes ($\mathcal{E} \leq 37$) have emerged from the scan (priority + TV21/TV10 secondary + isotopy variants): **mdecomp_48** (8d481a18 L1, scan $\mathcal{E} = 35$, **push best** $\mathcal{E} = 30$ ($cl_{13} = 14$, $cl_{23} = 16$, Session 21 trial 5!) — improving from $\mathcal{E} = 37 \rightarrow 33 \rightarrow 31 \rightarrow 30$ across trials; stable at $\mathcal{E} = 30$ across 25+ independent trials confirming a strong local basin), **mdecomp_71** (cd1785b9 L1, scan $\mathcal{E} = 37$, push best $\mathcal{E} = 31$ ($cl_{13} = 15$, $cl_{23} = 16$), stable across 16 trials then discontinued — strong local basin at $\mathcal{E} = 31$), **mdecomp_84** (cd1785b9 L1, scan $\mathcal{E} = 37$, push 4 trials all $\mathcal{E} = 38$, discontinued), **mdecomp_121** (6d7cf34c L1, scan $\mathcal{E} = 35$, push trials all $\mathcal{E} = 35$, discontinued), **mdecomp_142** (44aa4786 L1, scan $\mathcal{E} = 37$ ($cl_{13} = 19$, $cl_{23} = 18$), push 3 trials all $\mathcal{E} = 41$, discontinued), **mdecomp_163** (20d55e1d L1, scan $\mathcal{E} = 36$, $cl_{13} = 20$, $cl_{23} = 16$, push warmup + trial 1 both $\mathcal{E} = 40$, discontinued — redirected to mdecomp_191), **mdecomp_191** (bccfc7b0 L1, scan $\mathcal{E} = 36$, $cl_{13} = 23$, $cl_{23} = 13$, push: warmup+trials 1–2 all $\mathcal{E} = 39$, discontinued — stuck at $\mathcal{E} = 39$), **mdecomp_192** (20d55e1d L1, scan $\mathcal{E} = 36$, $cl_{13} = 23$, $cl_{23} = 13$), **sa_ct_212** (1343a5a6 L1, scan $\mathcal{E} = 37$, $cl_{13} = 16$, $cl_{23} = 21$), **mdecomp_217** (70f243cc L1, scan $\mathcal{E} = 36$, $cl_{13} = 19$, $cl_{23} = 17$), **sa_ct_236** (1343a5a6 L1, scan $\mathcal{E} = 37$, $cl_{13} = 20$, $cl_{23} = 17$), **sa_ct_290** (1343a5a6 L1, scan $\mathcal{E} = 35$, $cl_{13} = 16$, $cl_{23} = 19$), **mdecomp_279** (e54e791c L1, scan **E = 33**, $cl_{13} = 12$, $cl_{23} = 21$ — **co-scan-record!** (Session 25b); v3 scan-seed cold trial 1 finds $\mathcal{E} = 29$ ($cl_{13} = 12$, $cl_{23} = 17$)! **Session 29: hint cascade trial 4 (hint=E29) $\rightarrow$ E = 28** ($cl_{13} = 8$, $cl_{23} = 20$, seed=1046158808) — second independent (L1,L2) pair reaches $\mathcal{E} = 28$; trial 5 (hint=E28) running; trajectory $33 \rightarrow 29 \rightarrow 28$), and **mdecomp_281** (05f9f018 L1, scan **E = 33**, $cl_{13} = 12$, $cl_{23} = 21$; push v2 both $\mathcal{E} = 34$ (hint-anchored); **push v3 confirms $\mathcal{E} = 33$ basin:** cold trials 3,5 and hint trials 4,6 all $\mathcal{E} = 33$ ($cl_{13} = 20$, $cl_{23} = 13$) — 6 trials, all $\mathcal{E} = 33$; deep basin), and **mdecomp_296** (d42b3374 L1, scan $\mathcal{E} = 37$, $cl_{13} = 15$, $cl_{23} = 22$ — **10th distinct L1 family!**), and **mdecomp_45** (4cf608bd L1, scan $\mathcal{E} = 37$, $cl_{13} = 20$, $cl_{23} = 17$ — **11th distinct L1 family!**), and **iso_tv_21_0** (ee29e34e L1, scan **E = 32**, $cl_{13} = 13$, $cl_{23} = 19$ — **12th distinct L1 family!**, **all-time scan record!**; push v1 trials 1–5 all $\mathcal{E} = 32$; v2 trial 1 cold also $\mathcal{E} = 32$ (6 total, stable basin); v3 trial 1 cold $\mathcal{E} = 30$! ($cl_{13} = 15$, $cl_{23} = 15$); v3 trial 2 hint=E30 $\mathcal{E} = 29$! ($cl_{13} = 14$, $cl_{23} = 15$); v3 trial 3 cold $\mathcal{E} = 33$ (cold regression); **v3 trial 4 hint=E29 E = 26!!!** ($cl_{13} = 8$, $cl_{23} = 18$) — **NEW ALL-TIME RECORD!** 3σ below previous

best! trajectory $32 \rightarrow 30 \rightarrow 29 \rightarrow \mathbf{26}$; v3 trial 5 cold running, trial 6 hint=E26 targeting $\mathcal{E} \leq 20$), and **iso_tv_10_1** (557feb0f L1, scan $\mathcal{E} = 37$, $cl_{13} = 19$, $cl_{23} = 18$ — **13th distinct L1 family!**, TV10 isotopy variant; iso_tv_10_0 $\mathcal{E} = 40$, iso_tv_10_2 $\mathcal{E} = 43$), and **iso_mdecomp48_4** (1b41f88d L1, scan $\mathcal{E} = 37$, $cl_{13} = 16$, $cl_{23} = 21$ — **14th distinct L1 family!**, isotopy variant of mdecomp_48 (push record $\mathcal{E} = 30$); iso_mdecomp48_0-3,5 gave $\mathcal{E} = 38$ –41), and **iso2_tv21_0_1** (af53f0ef L1, scan $\mathcal{E} = 36$, $cl_{13} = 20$, $cl_{23} = 16$ — **15th distinct L1 family!**, isotopy variant of scan-record iso_tv_21_0; iso2 full scan: $E=36,40,42,37,38,40$ — 2 standouts out of 6 variants), and **iso2_tv21_0_3** (53a75a20 L1, scan $\mathcal{E} = 37$, $cl_{13} = 23$, $cl_{23} = 14$ — **16th distinct L1 family!**, isotopy variant of scan-record iso_tv_21_0), and **mdecomp_279** (e54e791c, see above — **17th distinct L1 family!**, scan $\mathcal{E} = 33$ co-record), and TV scan family **e0e09d80** (pairs tv_58,83,96,99,105 all $\mathcal{E} = 37$ — **18th distinct L1 family!**), and **tv_190** (8e540206 L1, scan $\mathcal{E} = 37$ — **19th distinct L1 family!**), and **tv_200** (17fc775b L1, scan $\mathcal{E} = 36$, $cl_{13} = 20$, $cl_{23} = 16$ — **20th distinct L1 family!**), and **mdecomp_216** (2b9a5282 L1, scan $\mathcal{E} = 36$, $cl_{13} = 19$, $cl_{23} = 17$ — **21st distinct L1 family!**), and **mdecomp_120** (752ef151 L1, scan $\mathcal{E} = 35$, $cl_{13} = 21$, $cl_{23} = 14$ — **22nd distinct L1 family!**, push trial starting next), and **mdecomp_289** (bbe685e3 L1, scan $\mathcal{E} = 35$, $cl_{13} = 16$, $cl_{23} = 19$ — **23rd distinct L1 family!**, push trial starting next), and **mdecomp_210** (37eebeb5 L1, scan $\mathcal{E} = 37$, $cl_{13} = 16$, $cl_{23} = 21$ — **24th distinct L1 family!**), and **mdecomp_235** (a79be19b L1, scan $\mathcal{E} = 37$, $cl_{13} = 20$, $cl_{23} = 17$ — **25th distinct L1 family!**), and **tv_230** (e0a8536d L1, scan $\mathcal{E} = 35$, $cl_{13} = 19$, $cl_{23} = 16$ — **26th distinct L1 family!**, TV21 secondary, scan continuation Session 26), and **tv_262** (31cd5484 L1, scan $\mathcal{E} = 36$, $cl_{13} = 15$, $cl_{23} = 21$ — **27th distinct L1 family!**, TV21 secondary), and **tv_223** (c5c54e19 L1, scan $\mathcal{E} = 37$ — **28th distinct L1 family!**, TV21 secondary). Session 21–26b: **iso_tv_21_0 is the scan record holder ($\mathcal{E} = 32$) and new push record holder ($\mathcal{E} = 26$)!** **iso_tv_21_0 v3 hint cascade:** $32 \rightarrow 30 \rightarrow 29 \rightarrow \mathbf{26}!!!$ (v1/v2: 6 trials stuck at $\mathcal{E} = 32$; v3 trial 1 cold $\mathcal{E} = 30$; trial 2 hint=E30 $\mathcal{E} = 29$; trial 3 cold $\mathcal{E} = 33$ (regression); trial 4 hint=E29 **E = 26!** **NEW ALL-TIME RECORD!**). Each hint trial improves on the previous best. Pool now 22 entries ($E=26$ at top). mdecomp_48 trajectory: $\mathcal{E} = 37 \rightarrow 33 \rightarrow 31 \rightarrow \mathbf{30}$ (trial 5, stable 25+ trials). **Scan complete: all 323/323 pairs scanned.** 28 distinct L1 families identified. New families from TV21 scan continuation: tv_230 (e0a8536d, $\mathcal{E} = 35$), tv_262 (31cd5484, $\mathcal{E} = 36$), tv_223 (c5c54e19, $\mathcal{E} = 37$). The 1343a5a6 family spans four pairs (sa_ct.212/236/286/290), scan $\mathcal{E} = 35$ –38; mdecomp_163/192 share L1=20d55e1d (both scan $\mathcal{E} = 36$); mdecomp_279/281/285: mdecomp_279 (e54e791c) scan $\mathcal{E} = 33$; mdecomp_281/285 share L1=05f9f018 (scan $\mathcal{E} = 33/38$). L1 family map: cd1785b9 (mdecomp_71, 84, 106), 8d481a18 (mdecomp_48), 6d7cf34c (mdecomp_121, 141), 44aa4786 (mdecomp_142), 20d55e1d (mdecomp_163, 192), bccfc7b0 (mdecomp_191, 162), 70f243cc (mdecomp_217), 1343a5a6 (sa_ct.212, 236, 286, 290), 05f9f018 (mdecomp_281, 285), d42b3374 (mdecomp_296), 4cf608bd (mdecomp_45), ee29e34e (iso_tv_21_0), 557feb0f (iso_tv_10_1), 1b41f88d (iso_mdecomp48_4), af53f0ef (iso2_tv21_0_1, $\mathcal{E} = 36$), 53a75a20 (iso2_tv21_0_3, $\mathcal{E} = 37$), e54e791c (mdecomp_279, **scan $\mathcal{E} = 33!$**), e0e09d80 (tv family, $\mathcal{E} = 37$), 8e540206 (tv_190, $\mathcal{E} = 37$), 17fc775b (tv_200, $\mathcal{E} = 36$), 2b9a5282 (mdecomp_216, scan $\mathcal{E} = 36$), 752ef151 (mdecomp_120, scan $\mathcal{E} = 35$), bbe685e3 (mdecomp_289, scan $\mathcal{E} = 35$), 37eebeb5 (mdecomp_210, scan $\mathcal{E} = 37$), a79be19b (mdecomp_235, scan $\mathcal{E} = 37$), e0a8536d (tv_230, scan $\mathcal{E} = 35$), 31cd5484 (tv_262, scan $\mathcal{E} = 36$), c5c54e19 (tv_223, scan $\mathcal{E} = 37$) — **twenty-eight distinct L1 hashes** in standout pairs, spanning $\mathcal{E} = 29$ –37 (note: af53f0ef, 53a75a20 isotopically equivalent to ee29e34e; e0e09d80 family spans 5+ TV pairs; **ee29e34e holds push record E = 26!**); identified across all **323/323 scanned pairs** (Sessions 17–26b; scan COMPLETE!). Note: 6 isotopy variants of mdecomp_48 scanned (all $\mathcal{E} = 37$ –41); 6 isotopy variants of scan-record iso_tv_21_0 scanned (iso2: $\mathcal{E} = 36$ –42, 2 standouts at $\mathcal{E} = 36, 37$). Push L strategy: mdecomp_48's $\mathcal{E} = 30$

is pair-specific; isotopy does not preserve push depth. **4-worker vs 1-worker CP-SAT:** iso.tv.21.0 was stuck at $\mathcal{E} = 32$ for 6 trials with 1-worker CP-SAT (v1: 5 trials, v2: 1 trial); the **4-worker v3 found $\mathcal{E} = 30$ in the very first cold trial**. This demonstrates that parallel CP-SAT workers (sharing learned clauses and exploring different search subtrees simultaneously) are qualitatively more powerful than single-worker for near-miss optimization. CPU strategy: 4-core machine fully dedicated. Currently running (Session 26b+): **iso.tv.21.0 v3** (trial 1 cold $\mathcal{E} = 30$; trial 2 hint $\mathcal{E} = 29$; trial 3 cold $\mathcal{E} = 33$; trial 4 hint=E29 $\mathcal{E} = 26$! NEW RECORD!; trial 5 cold $\mathcal{E} = 31$ (regression); **trial 6 hint=E26 running, seed=1014286829, targeting $\mathcal{E} \leq 20$**). mdecomp.279 discontinued (stable basin $\mathcal{E} = 35$ across 4 trials). Scan COMPLETE (all 323/323 done). Discontinued (stable basins, redirected compute): mdecomp.48 (25+ trials $\mathcal{E} = 30$), mdecomp.281 v3 (6 trials $\mathcal{E} = 33$), proof workers (24+ trials $\mathcal{E} = 28$ all TIMEOUT). mdecomp.45 discontinued (warmup $\mathcal{E} = 39$, outclassed by iso.tv.21.0). Is there a structural explanation for why both 74bb7b86 and 0968fc79 share the asymmetric distribution ($cl_{13} = 17$, $cl_{23} = 12$)?

5. Does the CT distribution (always ≤ 2) follow from a group-theoretic argument related to the factorization $10 = 2 \times 5$?
6. Why does the joint 3-LS SA converge to $\mathcal{E} = 37$? Is there a theoretical lower bound for the joint energy landscape below which only CP-SAT can descend?

8.3 Future Directions

1. **Wider CT search** with $>10,000$ pairs to test Conjecture 6.1.
2. **Isotopy diversification:** iso.tv.21.0 (an isotopy variant of TV21) gives scan $\mathcal{E} = 32$ and push $\mathcal{E} = 30$ — far better than the TV21 base pair ($\mathcal{E} \sim 37$). This pair now holds both the scan record and (jointly with mdecomp.48) the push record. Three isotopy variants of TV10 were scanned: iso.tv.10.1 (L1 557feb0f) yielded $\mathcal{E} = 37$ (13th distinct L1 family!), while iso.tv.10.0 gave $\mathcal{E} = 40$ and iso.tv.10.2 gave $\mathcal{E} = 43$. Future work should systematically generate isotopy transforms of all known promising base pairs (mdecomp.279/281, mdecomp.120/289, etc.) and push the best ones.
3. **Theoretical lower bound** on $\mathcal{E}$: can one prove $\mathcal{E} \geq 30$ (or higher) for all triples with $L_1 \perp L_2$ of order 10? The consistent SA barrier at $\mathcal{E} = 37$ across diverse starting points is suggestive but not a proof.
4. **Larger compute scale** (50+ workers on HPC cluster) for broader isotopy class coverage.
5. **Intercalate theory analysis** relating CT count to intercalate structure of MOLS-10 pairs (following Wanless & Webb [14]).
6. **Stronger CP-SAT strategies:** the current approach uses up to 4 parallel workers; deploying on HPC nodes with 32–128 workers per trial would be qualitatively stronger. Alternatively, Gurobi ILP or SDP-based bounds could give theoretical certificates on the minimum achievable $\mathcal{E}$.

A Representative MOLS-10 Pairs with Verification

This appendix gives full 10×10 matrices for three representative pairs from the catalog. For each pair, orthogonality is confirmed by verifying $cl(L_1, L_2) = 0$. Complete data for all 323 pairs is in `promising_pairs.json`.

A.1 Parker TV-21 (1960) — CT = 2

The canonical pair introduced by Parker [12]. This is the reference pair from which all **tv** and **ic*** variants derive. Verification: both squares are valid Latin squares; all 100 ordered symbol pairs appear exactly once in the superposition; $\text{cl}(L_1, L_2) = 0$; $\text{CT}(L_1, L_2) = 2$ (verified by exhaustive backtracking).

$$L_1^{\text{TV21}} = \begin{pmatrix} 0 & 9 & 4 & 7 & 6 & 5 & 3 & 8 & 2 & 1 \\ 9 & 5 & 0 & 2 & 3 & 8 & 6 & 4 & 1 & 7 \\ 2 & 6 & 9 & 0 & 4 & 3 & 1 & 7 & 5 & 8 \\ 4 & 1 & 6 & 9 & 8 & 2 & 7 & 0 & 3 & 5 \\ 7 & 8 & 5 & 1 & 2 & 0 & 4 & 9 & 6 & 3 \\ 1 & 7 & 3 & 4 & 5 & 6 & 8 & 2 & 0 & 9 \\ 8 & 3 & 7 & 5 & 1 & 9 & 2 & 6 & 4 & 0 \\ 6 & 0 & 2 & 3 & 9 & 7 & 5 & 1 & 8 & 4 \\ 5 & 4 & 8 & 6 & 7 & 1 & 0 & 3 & 9 & 2 \\ 3 & 2 & 1 & 8 & 0 & 4 & 9 & 5 & 7 & 6 \end{pmatrix} \quad L_2^{\text{TV21}} = \begin{pmatrix} 7 & 4 & 8 & 6 & 2 & 3 & 9 & 5 & 0 & 1 \\ 2 & 7 & 3 & 9 & 1 & 0 & 5 & 4 & 6 & 8 \\ 3 & 0 & 7 & 5 & 9 & 8 & 4 & 2 & 1 & 6 \\ 0 & 3 & 9 & 8 & 4 & 1 & 7 & 6 & 5 & 2 \\ 1 & 9 & 6 & 0 & 5 & 4 & 2 & 3 & 8 & 7 \\ 9 & 5 & 4 & 1 & 8 & 6 & 3 & 7 & 2 & 0 \\ 8 & 2 & 0 & 4 & 7 & 5 & 6 & 1 & 3 & 9 \\ 4 & 1 & 2 & 3 & 6 & 9 & 0 & 8 & 7 & 5 \\ 5 & 6 & 1 & 7 & 3 & 2 & 8 & 0 & 9 & 4 \\ 6 & 8 & 5 & 2 & 0 & 7 & 1 & 9 & 4 & 3 \end{pmatrix}$$

Orthogonality check: Superimposing L_1 and L_2 yields 100 cells; each of the 100 ordered pairs $(a, b) \in \{0, \dots, 9\}^2$ appears exactly once. $\text{cl}(L_1^{\text{TV21}}, L_2^{\text{TV21}}) = 0 \checkmark$.

A.2 TV-10 Variant — CT = 1 (new finding)

A transversal variant of the Parker L_1 , with a different orthogonal mate constructed by the **tv** method. This pair achieves $\text{CT} = 1$: there exists exactly one common transversal, which cannot form a partition of all 100 cells (10 disjoint CTs required for any valid L_3). By Corollary 2.3, no L_3 orthogonal to both squares exists. Verification: $\text{cl}(L_1, L_2) = 0$; $\text{CT} = 1$ (exhaustive enumeration).

$$L_1^{\text{TV10}} = \begin{pmatrix} 4 & 3 & 1 & 7 & 5 & 6 & 9 & 8 & 2 & 0 \\ 7 & 8 & 0 & 6 & 1 & 2 & 4 & 9 & 3 & 5 \\ 8 & 6 & 4 & 1 & 9 & 5 & 3 & 7 & 0 & 2 \\ 5 & 9 & 6 & 3 & 4 & 0 & 1 & 2 & 7 & 8 \\ 2 & 0 & 7 & 5 & 6 & 4 & 8 & 1 & 9 & 3 \\ 1 & 7 & 9 & 0 & 8 & 3 & 2 & 5 & 6 & 4 \\ 9 & 5 & 8 & 4 & 2 & 7 & 0 & 3 & 1 & 6 \\ 6 & 2 & 3 & 9 & 0 & 1 & 5 & 4 & 8 & 7 \\ 3 & 4 & 2 & 8 & 7 & 9 & 6 & 0 & 5 & 1 \\ 0 & 1 & 5 & 2 & 3 & 8 & 7 & 6 & 4 & 9 \end{pmatrix} \quad L_2^{\text{TV10}} = \begin{pmatrix} 3 & 8 & 0 & 5 & 1 & 2 & 4 & 7 & 9 & 6 \\ 9 & 4 & 5 & 1 & 8 & 6 & 7 & 2 & 3 & 0 \\ 2 & 0 & 4 & 7 & 3 & 9 & 5 & 6 & 1 & 8 \\ 4 & 5 & 6 & 2 & 0 & 7 & 9 & 1 & 8 & 3 \\ 0 & 9 & 2 & 3 & 4 & 8 & 1 & 5 & 6 & 7 \\ 1 & 3 & 9 & 0 & 6 & 4 & 2 & 8 & 7 & 5 \\ 7 & 2 & 8 & 6 & 5 & 1 & 3 & 0 & 4 & 9 \\ 5 & 7 & 1 & 8 & 2 & 3 & 6 & 9 & 0 & 4 \\ 6 & 1 & 3 & 9 & 7 & 0 & 8 & 4 & 5 & 2 \\ 8 & 6 & 7 & 4 & 9 & 5 & 0 & 3 & 2 & 1 \end{pmatrix}$$

Note: $\text{CT} = 1$ is the minimum possible for a MOLS pair. The existence of such pairs with $\text{CT} = 1$ is itself a new finding; prior to this work no systematic CT enumeration of MOLS-10 pairs had been published.

A.3 Transversal Decomposition Pair mdecomp_48 — CT = 2 (new construction)

Generated by the **mdecomp** method: a random Latin square L_1 is constructed (not isotopic to Parker's), then a transversal decomposition finds an orthogonal mate L_2 . This pair and all 123 other **mdecomp** pairs have L_1 matrices outside the Parker isotopy class, making them genuinely independent new MOLS-10 pairs. Verification: $\text{cl}(L_1, L_2) = 0$; $\text{CT} = 2$.

$$L_1^{\text{md48}} = \begin{pmatrix} 8 & 1 & 4 & 2 & 6 & 3 & 0 & 5 & 9 & 7 \\ 4 & 6 & 9 & 0 & 8 & 5 & 2 & 7 & 1 & 3 \\ 3 & 7 & 5 & 6 & 2 & 4 & 8 & 9 & 0 & 1 \\ 5 & 9 & 1 & 3 & 7 & 8 & 6 & 0 & 2 & 4 \\ 7 & 8 & 0 & 4 & 1 & 2 & 9 & 3 & 5 & 6 \\ 9 & 5 & 2 & 8 & 3 & 1 & 4 & 6 & 7 & 0 \\ 6 & 4 & 3 & 1 & 9 & 0 & 7 & 2 & 8 & 5 \\ 2 & 3 & 7 & 5 & 0 & 6 & 1 & 8 & 4 & 9 \\ 0 & 2 & 6 & 7 & 4 & 9 & 5 & 1 & 3 & 8 \\ 1 & 0 & 8 & 9 & 5 & 7 & 3 & 4 & 6 & 2 \end{pmatrix} \quad L_2^{\text{md48}} = \begin{pmatrix} 8 & 9 & 5 & 2 & 0 & 4 & 1 & 6 & 3 & 7 \\ 9 & 2 & 8 & 7 & 3 & 1 & 5 & 0 & 4 & 6 \\ 2 & 8 & 7 & 3 & 6 & 0 & 4 & 5 & 9 & 1 \\ 3 & 6 & 0 & 8 & 1 & 5 & 9 & 2 & 7 & 4 \\ 4 & 7 & 6 & 1 & 8 & 3 & 0 & 9 & 2 & 5 \\ 1 & 4 & 9 & 0 & 7 & 2 & 6 & 8 & 5 & 3 \\ 7 & 3 & 1 & 5 & 9 & 8 & 2 & 4 & 6 & 0 \\ 0 & 5 & 3 & 9 & 4 & 6 & 7 & 1 & 8 & 2 \\ 5 & 1 & 4 & 6 & 2 & 7 & 8 & 3 & 0 & 9 \\ 6 & 0 & 2 & 4 & 5 & 9 & 3 & 7 & 1 & 8 \end{pmatrix}$$

Orthogonality check: $\text{cl}(L_1^{\text{md48}}, L_2^{\text{md48}}) = 0 \checkmark$. $\text{CT} = 2$. This pair's L_1 has first row $[8, 1, 4, 2, 6, 3, 0, 5, 9, 7]$, confirming it is not isotopic to TV-21's L_1 (which has first row $[0, 9, 4, 7, 6, 5, 3, 8, 2, 1]$ in reduced form).

A.4 Complete 311-Pair Summary Table

Construction	Total	CT=1	CT=2
tv (Parker transversal variants)	56	46	10
ic2 (intercalate chain, depth 2)	3	1	2
ic3 (intercalate chain, depth 3)	3	1	2
ic5 (intercalate chain, depth 5)	5	2	3
ic8 (intercalate chain, depth 8)	4	1	3
ic12 (intercalate chain, depth 12)	3	0	3
ic1.3 (L1 intercalate, depth 3)	2	1	1
ic1.5 (L1 intercalate, depth 5)	2	1	1
iso (isotopy transforms)	6	3	3
mdecomp (transversal decomposition)	124	2	122
sa_ct (SA CT-climbing)	103	0	103
Total	311	58	253

All 311 pairs are verified MOLS ($\text{cl} = 0$) with CT computed by exhaustive backtracking. **No pair has** $\text{CT} > 2$ — every pair is SAT-certified to have no orthogonal mate L_3 (Glucose3 CDCL, average 42 ms/pair, 0 timeouts).

B Data Tables

Sample E=37 Near-Miss Triple

One of the five $\mathcal{E} = 37$ triples (pool entry 0, found 2026-06-07 18:41). All matrices have symbols $\{0, \dots, 9\}$.

The full triple $(L_1^{(0)}, L_2^{(0)}, L_3^{(0)})$ from pool entry 0. $L_1^{(0)} \perp L_2^{(0)}$ is verified ($\text{cl}_{12} = 0$); $L_3^{(0)}$ is a valid Latin square with $\text{cl}_{13} = 22$, $\text{cl}_{23} = 15$.

$$L_1^{(0)} = \begin{pmatrix} 4 & 7 & 3 & 2 & 8 & 9 & 5 & 1 & 0 & 6 \\ 8 & 2 & 5 & 1 & 0 & 4 & 7 & 6 & 3 & 9 \\ 1 & 0 & 9 & 3 & 4 & 8 & 6 & 2 & 7 & 5 \\ 7 & 3 & 6 & 4 & 5 & 0 & 8 & 9 & 2 & 1 \\ 9 & 8 & 4 & 7 & 6 & 1 & 2 & 0 & 5 & 3 \\ 3 & 1 & 0 & 5 & 7 & 6 & 9 & 8 & 4 & 2 \\ 0 & 4 & 1 & 6 & 2 & 5 & 3 & 7 & 9 & 8 \\ 6 & 5 & 2 & 8 & 9 & 7 & 0 & 3 & 1 & 4 \\ 5 & 6 & 7 & 9 & 3 & 2 & 1 & 4 & 8 & 0 \\ 2 & 9 & 8 & 0 & 1 & 3 & 4 & 5 & 6 & 7 \end{pmatrix} \quad L_2^{(0)} = \begin{pmatrix} 6 & 0 & 2 & 3 & 5 & 9 & 7 & 1 & 4 & 8 \\ 3 & 4 & 6 & 7 & 8 & 5 & 1 & 0 & 9 & 2 \\ 2 & 3 & 5 & 1 & 0 & 6 & 9 & 8 & 7 & 4 \\ 9 & 8 & 7 & 4 & 2 & 1 & 0 & 3 & 5 & 6 \\ 8 & 7 & 9 & 2 & 1 & 4 & 6 & 5 & 3 & 0 \\ 5 & 9 & 0 & 8 & 6 & 3 & 4 & 2 & 1 & 7 \\ 7 & 2 & 8 & 5 & 9 & 0 & 3 & 4 & 6 & 1 \\ 4 & 5 & 1 & 9 & 7 & 8 & 2 & 6 & 0 & 3 \\ 1 & 6 & 3 & 0 & 4 & 2 & 5 & 7 & 8 & 9 \\ 0 & 1 & 4 & 6 & 3 & 7 & 8 & 9 & 2 & 5 \end{pmatrix}$$

$$L_3^{(0)} = \begin{pmatrix} 6 & 1 & 2 & 0 & 5 & 7 & 3 & 9 & 4 & 8 \\ 8 & 9 & 0 & 1 & 6 & 3 & 2 & 7 & 5 & 4 \\ 4 & 2 & 3 & 7 & 8 & 1 & 0 & 5 & 6 & 9 \\ 9 & 3 & 7 & 5 & 1 & 6 & 4 & 8 & 2 & 0 \\ 2 & 0 & 5 & 8 & 3 & 4 & 7 & 1 & 9 & 6 \\ 3 & 8 & 9 & 4 & 7 & 5 & 1 & 6 & 0 & 2 \\ 0 & 7 & 1 & 9 & 4 & 2 & 6 & 3 & 8 & 5 \\ 7 & 6 & 8 & 2 & 9 & 0 & 5 & 4 & 3 & 1 \\ 1 & 5 & 4 & 6 & 0 & 9 & 8 & 2 & 7 & 3 \\ 5 & 4 & 6 & 3 & 2 & 8 & 9 & 0 & 1 & 7 \end{pmatrix}$$

Verified: $L_1^{(0)} \perp L_2^{(0)}$ ($\text{cl}_{12} = 0$); $L_3^{(0)}$ is a valid Latin square; $\text{cl}_{13} = 22$, $\text{cl}_{23} = 15$, $\mathcal{E} = 37$. Full data for all five $\mathcal{E} = 37$ triples is in `mols10/results/near_miss_triple.json`.

Notation Summary

Symbol	Meaning
$N(n)$	Max number of MOLS of order n
L_i	Latin square i (10×10 , symbols 0–9)
$L_i \perp L_j$	L_i orthogonal to L_j
$\text{cl}(A, B)$	Clash count: symbol pairs missing from superposition
$\mathcal{E}$	Total energy = $\text{cl}_{12} + \text{cl}_{13} + \text{cl}_{23}$
$\text{CT}(L_1, L_2)$	Common transversal count
SA	Simulated annealing
PT	Parallel tempering
ILS	Iterated local search
CDCL	Conflict-driven clause learning

Data availability. All near-miss triples, MOLS pairs, worker logs, and search code are available at <https://github.com/ajzhanghk/autoresearch-glm>, branch `claude/mols-order-10-search-yfQXK`.

References

- [1] Audemard, G., & Simon, L. (2009). Predicting learnt clauses quality in modern SAT solvers. *Proceedings of IJCAI*, 399–404.
- [2] Google OR-Tools Team. (2023). OR-Tools: Google’s operations research tools. <https://developers.google.com/optimization>. Version 9.x, CP-SAT solver.
- [3] Ignatiev, A., Morgado, A., & Marques-Silva, J. (2019). RC2: An efficient MaxSAT solver. *Journal on Satisfiability, Boolean Modeling and Computation*, 11(1), 53–64.
- [4] Clements, G. F., & Lindström, B. (1965). A generalization of a combinatorial theorem of Macaulay. *Journal of Combinatorial Theory*, 7(3), 230–238.
- [5] Colbourn, C. J., & Dinitz, J. H. (Eds.) (2006). *Handbook of Combinatorial Designs* (2nd ed.). Chapman & Hall/CRC.
- [6] Dénes, J., & Keedwell, A. D. (1991). *Latin Squares: New Developments in the Theory and Applications*. North-Holland.
- [7] Euler, L. (1782). Recherches sur une nouvelle espèce de quarrés magiques. *Verhandelingen uitgegeven door het zeeuwisch genootschap der wetenschappen te Vlissingen*, 9, 85–239.

- [8] Geyer, C. J. (1991). Markov chain Monte Carlo maximum likelihood. *Proceedings of the 23rd Symposium on the Interface*, 156–163.
- [9] Knuth, D. E. (2000). Dancing links. *Millennial Perspectives in Computer Science*, 187–214.
- [10] Lam, C. W. H., Thiel, L., & Swiercz, S. (1989). The nonexistence of finite projective planes of order 10. *Canadian Journal of Mathematics*, 41(6), 1117–1123.
- [11] McKay, B. D., Meynert, A., & Myrvold, W. (2007). Small Latin squares, quasigroups, and loops. *Journal of Combinatorial Designs*, 15(2), 98–119.
- [12] Parker, E. T. (1960). Orthogonal Latin squares. *Proceedings of the National Academy of Sciences*, 47(6), 859–862.
- [13] Tarry, G. (1901). Le problème des 36 officiers. *Compte Rendu de l'Association Française pour l'Avancement des Sciences Naturelles*, 2, 170–203.
- [14] Wanless, I. M., & Webb, B. S. (2011). The existence of Latin squares without orthogonal mates. *Designs, Codes and Cryptography*, 60(2), 143–151.

Table 3: Best energy $\mathcal{E}$ per session.

Session	Key development
1	Initial 4-replica PT
2	Intercalate moves added
3	Pool warm-starts; kscramble
4	6-replica PT; bigscramble
5	7-replica PT; time-based ILS
6	D5/affine/cyclic inits; L3 dedup fix
7	SAT worker; pool-jump ILS
8	Crossover, MAX-SAT, row-replace, joint 3-LS SA
9	CP-SAT optimizer + symbreak; E=34 for 2 independent pairs
10	SA/CP-SAT cascade; file-locking fix; E=32 via rapid descent
11	Long-duration 8w (900s) CP-SAT breaks to E=31 ; cascade 32 $\rightarrow$ 31
12	L3-only SA (cell-swap bug found/fixed); 12 parallel 900s CP-SAT sweeps; E=31 holds
13	Second distinct E=31 pair found (joint SA); proof workers probe $\mathcal{E} \leq 30$; E=31 holds
14	Warm-start bug fixed in proof worker; $\mathcal{E} = 30$ found in 311s; isotopy: all 7 E=31 pairs same class
14b	2-move intercalate neighborhood search on E=30 finds $\mathcal{E} = 29$ instantly (cl ₁₃ =15, cl ₂₃ =14)
14c	5-move local min confirmed (22.2M quintis); 6 diff-class E=31 pairs stuck at E=31; E=32 entry (cla
15	Three new classes (3a33165a, 74bb7b86, 0968fc79) reach E=30 in 1200s deep opt; 12 proof trials (I
16	Push-to-E29: 3a33165a E=31 $\rightarrow$ 30 (1800s), E $\leq$ 29 TIMEOUT; 74bb7b86 E=31$\rightarrow$29 (cl ₁₃ = 17, c
17	Trials 1–2: all 4 main + cold TIMEOUT ($\times 5$); push 3a33165a 2 $\times$ E=31 $\rightarrow$ 31 (no gain, discontinued
18	Pair scan reaches mdecomp pairs; mdecomp_48 (new class 8d481a18) achieves E=35 in 180
19	mdecomp_71 (new class d5a580f8) E=37 in 180s , then E=35 warmup with 2 workers ; r
20	Scan reaches 37/247 priority pairs; mdecomp_84 (3rd notable pair, E=37) ; push trial 1 (1800
21	mdecomp_48 reaches $\mathcal{E} = 30$ (cl ₁₃ = 14, cl ₂₃ = 16, trial 5!) — one step from $\mathcal{E} = 29$; mdecomp_7
22	Scan record $\mathcal{E} = 33$ (mdecomp_281, L1=05f9f018, cl ₁₃ = 12, cl ₂₃ = 21); all 227 priority pairs s
23	NEW SCAN RECORD $\mathcal{E} = 32$ (iso_tv_21_0, L1=ee29e34e, cl ₁₃ = 13, cl ₂₃ = 19) — 12th L1 fam
24	Container reset : all workers restarted from saved best-L3 files; iso_tv_21_0 push resumed trial 5 (
25	iso_mdecomp48 scan : 6 isotopy variants, all $\mathcal{E} = 37$ –41 (best: iso_mdecomp48_4 E=37, 14th L1 f
25b	mdecomp_279 discovered as co-scan-record ($\mathcal{E} = 33$, e54e791c, 17th L1 family, first push!); 5
26	iso_tv_21_0 v3 trial 2 hint=E30 $\mathcal{E} = 29!$ (cl ₁₃ = 14, cl ₂₃ = 15, seed=1333021984) — SECONI
26b	iso_tv_21_0 v3 trial 4 hint=E29 $\mathcal{E} = 26!!!$ (cl ₁₃ = 8, cl ₂₃ = 18, seed=1943590465) — NEW A
27	Dual-strategy enhanced push : (1) v5 short-timeout (300s, 2 workers, hint=E26) for 3 $\times$ mor
28	Local-minimum analysis + smart multi-family attack : 1-move neighborhood analysis proves
29	Three-front attack continues: mdecomp_279 hint cascade (trial 4, hint=E29) achieves E =
30	New-pair scan + mdecomp_113 breakthrough : e29 cascade killed (12 trials, aba463bd/0c3b8

Table 4: MOLS-10 pair catalog: breakdown by construction method.

Construction method	Pairs	CT=1	CT=2
TV variants (Parker 1960 transversals)	56	46	10
Intercalate chains (depths 2–12)	22	8	14
Isotopy transforms	6	3	3
Transversal decomposition (mdecomp)	124	2	122
SA CT-climbing (sa_ct)	103	0	103
Total	311	58	253

Table 5: CT count distribution across all 311 MOLS-10 pairs.

CT value	Pairs	Fraction
1	58	18.4%
2	258	81.6%
≥ 3	0	0%

Table 6: SAT worker cumulative results (Session 8).

Metric	Phase 1	Phase 2
Pairs tested	311	ongoing
SAT (L3 found)	0	0
UNSAT (certified)	311	ongoing
Timeout	0	0
Average solve time	42 ms	40 ms

Table 7: Near-miss pool state (Session 16, top entries). Six E=29 solutions found across four distinct isotopy classes; one class at E=30.

Index	Isotopy class	Solver	$\mathcal{E}$	cl ₁₂	cl ₁₃	cl ₂₃
0	4f04a773	2-move nbhd from E=30	29	0	15	14
1	4f04a773	CP-SAT alt-pair sweep	29	0	15	14
2	4f04a773	CP-SAT perturb sweep	29	0	15	14
3	2cc8e22f	CP-SAT opt. from E=32	29	0	14	15
4	74bb7b86	CP-SAT build-and-solve (1800 s)	29	0	17	12
5	0968fc79	CP-SAT build-and-solve (1800 s)	29	0	17	12
6	4f04a773	CP-SAT proof (311 s)	30	0	17	13
7–9	d83f1182, 48db40d4, ebfeb9f7	CP-SAT opt. (1200 s)	31	0	19	12
10	3a33165a	CP-SAT opt. (1800 s)	31→30	0	~15	~15